NOTES OF
Star Formation Physics

version 1.0

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For those who come after

PREFACE

> *In a sense, you might say I'm getting quite fond of writing these kind of notes.*

Not long ago, it occurred to me how cool it is when someone unexpectedly releases a very detailed and all-comprehensive version of their notes, especially when dealing with a course that has a handful of topics often interacting together in unpredictable, yet fascinating, ways.

These notes will be mainly based on *my* own notes of the lectures by Professor F. Fontani and M. Padovani.

These notes refer to the lectures of the academic year 2026-2027, so future versions of these lectures might differ slightly. However, since I take little to no pride in my messy notes, I'll be often using some of the references you can find on the course catalogue page or that have been cited or shown during class. Either way, I'll do my best keeping track of them all in the bibliography of this humble collection.

You can report errors (whatever their nature might be) and suggestions for additions at g.pannocchia3@studenti.unipi.it or through whatever convoluted way (conventional or not) you prefer[1].

You can find all the notes I've redacted here.

Without further ado, we'd better not lose much more time on a preface and get started with it.

> *There was Eru, the One, who in Arda is called Ilúvatar; and he made first the Ainur* [...] *But for a long while they sang only each alone, or but few together, while the rest hearkened; for each comprehended only that part of the mind of Ilúvatar from which he came, and in the understanding of their brethren they grew but slowly.*
> *Yet ever as they listened they came to deeper understanding, and increased in unison and harmony.*
>
> *Ainulindalë*, "The music of the Ainur",
> Silmarillion, J. R. R. Tolkien

1 I'd like, however, not to see my house stormed by homing pidgeons.

CONTENTS

1 The Interstellar Medium 1
- 1.1 Collisional Processes 4
  - 1.1.1 Collisional Rate Coefficients 4
- 1.2 Statistical Mechanics 5
  - 1.2.1 Partition Function and Levels Population 5
- 1.3 Energy Levels of Molecules 7
  - 1.3.1 Lines Hyperfine Structure 10
  - 1.3.2 Inversion Transitions of NH_3 10

2 Introduction to Astrochemistry 13
- 2.1 Surface Chemistry 13
- 2.2 Gas-Phase Chemistry 14
  - 2.2.1 Neutral-Neutral Reactions 15
  - 2.2.2 Neutral-Charged Reactions 15
- 2.3 Formation of H_3^+ 15
- 2.4 Formation and Destruction of CO 16
- 2.5 Freeze-Out In Collapsing Clouds 17
- 2.6 Isotopic Fractionation 18

3 Radiative Transfer 19
- 3.1 Introduction 19
- 3.2 Relevant quantities for radiative transfer 19
- 3.3 Blackbody Function 21
- 3.4 Radiative transfer equation 24
  - 3.4.1 Monochromatic emission coefficient 25
  - 3.4.2 Absorption coefficient 25
- 3.5 Kirchhoff's Law and LTE 27
- 3.6 Einstein Coefficients and Line Profile 30
  - 3.6.1 Line Broadening 32
- 3.7 Relations between the Einstein Coefficients 34
  - 3.7.1 Absorption and Emission coefficients in terms of Einstein coefficients 35
- 3.8 Statistical Equilibrium 36
  - 3.8.1 Maser lines 37
- 3.9 Excitation Temperature and Column Density 39
  - 3.9.1 Boltzmann Diagrams 40
- 3.10 nLTE and Rate of Collisional Transitions 41
- 3.11 Observations of Molecular Clouds and SF Regions 43
- 3.12 Optical Depth Measurements 46
- 3.13 Measurements of Densities and Masses 47
  - 3.13.1 Density of Molecular Hydrogen 48
  - 3.13.2 Dust and Gas Mass 48

Bibliography 53

1 THE INTERSTELLAR MEDIUM

For this part of this chapter, I'll be mainly following [5], Chapters 1-2, but I won't fail to report other used references were I to do so.

The interstellar medium, or ISM, is, arguably, also the most important component of galaxies, for it is the ISM that is responsible for forming the stars that are the dominant sources of energy. While it now appears that the mass of most galaxies is primarily in the form of dark matter particles that are collisionless, or nearly so, it is the baryons (accounting for perhaps $\sim 10\%$ of the total mass) that determine the visible appearance of galaxies, and that are responsible for nearly all of the energy emitted by galaxies, derived from nuclear fusion in stars and the release of gravitational energy in accretion disks around black holes.

The ISM has mainly four constituents:

1. Gas, which covers a whooping 99% of the total mass of the ISM. Note that when we refer to gas, we're imagining it as a macroscopic fluid/-body;

2. Dust , which has a significantly lower density than gas $\rho_d \sim 1\%\rho_g$. Dust is made of a hefty amount of *refractory elements*, mainly coming in the form of grains of different morphology and diameters in the range $D \sim 0.1 - 10\,\mu$m. Dust is mostly present in cold environments, but incidentally also in regions where $T \sim 10^5 - 10^6$;

   With the term refractory elements we're referring to solid particles that evaporate/sublimate at temperatures $T \gg 1000\,K$. For the sake of simplicity, they're usually assumed to be spherical.

3. Cosmic Rays are charged, highly energetic particles produced in extreme environments, like supernovae and the stellar winds of O/B-class stars. The vast majority of CR are protons ($\sim 90\%$), followed by α particles and electrons. Both have velocities much greater than those typical of the ISM: The most energetic ever detected had an energy of $\sim 10^{21}$ eV. As such, they can easily penetrate even the most dense regions of the ISM.

4. Photons from many sources, including the cosmic microwave background (CMB) in the millimeter/radio band, stellar photospheres (i.e., starlight) in the optical band, radiation emitted by interstellar ions, atoms, and molecules in the infrared band; thermal emission from interstellar grains that have been heated by starlight still in the infrared. Then moving to the higher energy (γ and X) we also observe photons from free-free emission ("Bremsstrahlung") from interstellar plasma, synchrotron radiation from relativistic electrons, and gamma rays emitted in nuclear transitions and π_0 decays.

The mass of the gas component alone makes about 10% of the total mass of a galaxy, but it still is lower than the mass "stored" through stars. That said,

the ISM is far more spatially extended than stars ($\sim 25 - 27\,\mathrm{kpc}$). Typical conversion rates gas$\longleftrightarrow$stars are

$$\frac{\mathrm{d}M}{\mathrm{d}t}(\text{gas} \rightarrow \text{stars}) \sim 1 - 5 M_\odot\,\mathrm{yr}^{-1} \quad \frac{\mathrm{d}M}{\mathrm{d}t}(\text{stars} \rightarrow \text{gas}) \sim 1 M_\odot\,\mathrm{yr}^{-1}$$

The baryons in the interstellar medium of the Milky Way are found with a wide range of temperatures and densities; because the interstellar medium is dynamic, all densities and temperatures within these ranges can be found somewhere in the Milky Way. However, it is observed that most of the baryons have temperatures falling close to various characteristic states, or "phases." For purposes of discus- sion, it is convenient to name these phases. Here we identify seven distinct phases[1] that, between them, account for most of the mass and most of the volume of the interstellar medium. The seven phases are defined based on the temperature, density, and ionization status of Hydrogen atoms. These phases consist of the following:

1. *Hot ionized medium* (HIM) or *Coronal Gas*: Gas that has been shock-heated to temperatures $T > 10^6\,\mathrm{K}$ by blastwaves racing outward from supernova explosions. The gas is collisionally ionized, with ions such as OVI (i.e. O^{5+}) present. Most of the coronal gas has low density ($n < 0.01\,\mathrm{cm}^{-3}$), filling an appreciable fraction ($10 - 50\%$) of the volume of the galactic disk. The coronal gas regions may have characteristic dimensions of $\sim 20\,\mathrm{pc}$, and may be connected to other coronal gas volumes. The coronal gas cools on $\sim$ Myr time scales. Much of the volume above and below the disk is thought to be pervaded by coronal gas. The main heating mechanisms are (in individual galaxies) supernovae blast waves, (in galaxy clusters) release of gravitational energy, while they cool through X/γ-rays emission. The main tracers for this region are X/γ/UV-rays emission and radio emission from synchrotron radiation;

The Sun is located in the (warm) Local Bubble, a cavity with diameter of $\sim 300\,lyr$, which could have been created by the esplosion of a nearby Supernova (Geminga),$300,000\,yr$ ago

2. *Warm Ionized* H_{II} *Medium*: Gas where the hydrogen has been photoionized by ultraviolet photons from hot stars. Most of this photoionized gas is maintained by radiation from recently formed hot massive O-type stars - the photoionized gas may be dense material from a nearby cloud (in which case the ionized gas is called an H_{II} region) or lower density "intercloud" medium (referred to as diffuse H_{II}). The density of the WIM depends on the particular environment (but roughly $n \sim 1\,\mathrm{cm}^{-3}$), but has a temperature of about $T \sim 10^4 - 10^5\,\mathrm{K}$ and fills about $10 - 15\%$ of the volume of the galactic disk. The main heating mechanism is UV radiation from O-B stars, while they cool through high-excitation transitions of atoms. The main tracers for this region are radio lines and continuum (recombination lines, free-free), optical/UV lines;

3. *Warm neutral medium* (WNM): Predominantly atomic gas heated to temperatures $T \approx 10^{3.7}\,\mathrm{K}$. In the local interstellar medium, this gas is found at densities $n \approx 0.6\,\mathrm{cm}^{-3}$. It fills a significant fraction of the volume of the disk, perhaps 40%. The main heating mechanism is

1 Although how well-separated these phases are is a completely different question.

photoelectric effect on grains by photons, while cooling occurs mainly through the 21 cm line of atomic Hydrogen or through the 158 μm line of ionized Carbon, which are also the main tracers;

4. *Cold neutral medium* (CNM): Predominantly atomic gas at temperatures $T \approx 10^2\,\mathrm{K}$, with densities $n \approx 30\,\mathrm{cm}^{-3}$ filling $\sim 1\%$ of the volume of the local interstellar medium. The main heating mechanisms are the interaction with cosmic rays and photoelectric effect on grains by photons, while cooling occurs mainly through the 21 cm line of atomic Hydrogen, which is also the main tracer;

5. *Molecular gas*: Two different phases can be distinguished here: the *diffuse molecular gas* and the *dense molecular gas*.
   - *Diffuse molecular gas*: similar to the cool H_I clouds, but with sufficiently large (column) densities so that the self-shielding of H_2 allows H_2 molecules to be abundant in the cloud interior. It has a temperature of $T \sim 50\,\mathrm{K}$ and a density $n \sim 100\,\mathrm{cm}^{-3}$ and fills about 0.1% of the total galactic volume. The main heating mechanism is the interaction with cosmic rays, while cooling occurs mainly through the lines of atomic metals. The main tracer is molecular Hydrogen (but in absorption alone);
   - *Dense molecular gas*: Gravitationally bound clouds achieving $n > 100\,\mathrm{cm}^{-3}$. These clouds are often "dark" - with visual extinction $A_V \gtrsim 3\,\mathrm{mag}$ through their central regions. In these dark clouds, the dust grains are often coated with "mantles" composed of H_2O and other molecular ices. It is within these regions that star formation takes place. They fill only the 0.001% of the total galactic volume, but make about 1/4 of the total mass. The main heating mechanism is the interaction with cosmic rays, while cooling occurs mainly through the roto-vibrational molecular lines of molecules, which are also the main tracers.

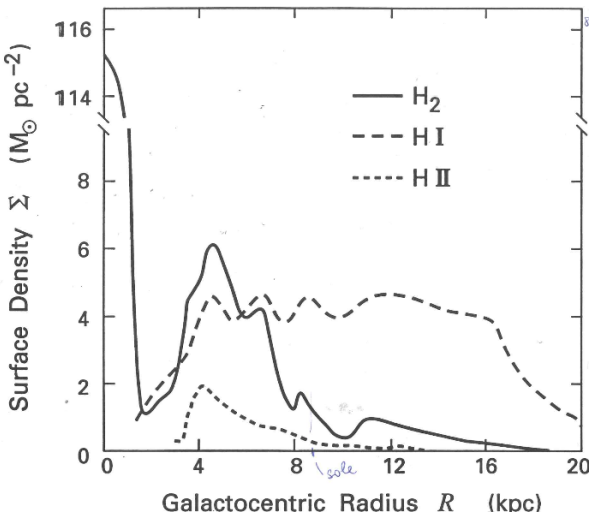


Figure 1.1: Distribution of different ionization states of Hydrogen as a function of the galactocentric radius. Credits: Unknown.

1.1 COLLISIONAL PROCESSES

Collisions are fundamental to the physics of the interstellar medium (ISM): They allow the gas to (usually) be treated as a fluid; they determine the thermal and electrical conductivity and diffusion coefficients; they produce most of the excitations of ions, atoms, and molecules that result in emission of photons from the ISM; and they are responsible for recombination of electrons and ions, and for chemical reactions.

1.1.1 Collisional Rate Coefficients

The rate per unit volume of a general two-body collisional process

$$\mathrm{A} + \mathrm{B} \rightarrow \text{products}$$

is written as

$$r_{AB} = n_A n_B \langle \sigma v \rangle_{AB} \tag{1.1}$$

where the two-body collisional rate coefficient for the general collisional process is

$$\langle \sigma v \rangle_{AB} = \int_0^\infty \sigma_{AB}(v) v f_f \, \mathrm{d}v \tag{1.2}$$

where σ_{AB} is the velocity velocity-dependent reaction cross section for the reaction and we're allowing the individual particles to have a non-trivial velocity distribution $f_v \, \mathrm{d}v$. The collision rate coefficient thus includes two contributions: An "internal" part that takes into account both the phyisico/-chemical properties of the colliders, and the quantum probabilities that a given interaction will produce a given product, and an "external" part that takes into account the kinetical status (i.e. v) of colliders.

If we assume thermal equilibrium at temperature T, the distribution function for the relative speed in encounters between particles A and B is given by a Maxwellian velocity distribution

$$f_v \, \mathrm{d}v = 4\pi \left(\frac{\mu}{2\pi k T} \right)^{3/2} \exp\left(-\frac{\mu v^2}{2kT} \right) v^2 \, \mathrm{d}v \tag{1.3}$$

where μ is the reduced mass of the collision partners. Note that if the velocity distribution of the individual collision partners is a Maxwellian distribution, the (relative) velocity distribution is still a Maxwellian and has the form (1.3).

Let us give a brief overview of the collisional rate coefficient for different types of interaction.

Interaction between neutral species

The main forces at play between atoms, grains, and neutral molecules are long-range forces (attractive forces between istantaneous dipoles) and close-range forces (repulsive Coulombian forces of electron). This configuration is easily approximated by the interaction between rigid spheres, allowing us to find an expression with the likes of

$$\langle \sigma v \rangle_{nn} = 1.81 \cdot 10^{-10} \left(\frac{T}{100\,\mathrm{K}} \right)^{1/2} \left(\frac{m_\mathrm{H}}{\mu} \right)^{1/2} \left(\frac{r_A + r_B}{2\,\text{Å}} \right)^2 \mathrm{cm}^{-3}\,\mathrm{s}^{-1} \tag{1.4}$$

Interaction between neutral and charged species

This is the case of an atom or neutral molecule interacting with an ion or electron. The relevant forces at play here are the attractive dipolar forces, which are non-negligible even at long distances. Unfortunately, this means that the rigid sphere approximation is no longer valid.

We should then consider the force exerted by the ion on the induced dipole

$$F = -\frac{2\alpha Z^2 e^2}{r^5} \tag{1.5}$$

to which is associated an interaction potential

$$U(r) = -\frac{1}{2}\frac{\alpha Z^2 e^2}{r^4} \tag{1.6}$$

A proper discussion should then consider the impact parameter for the process and deduce from there the collisional rate coefficient, which we'll here present without proof

$$\langle \sigma v \rangle_{nc} = 8.98 \cdot 10^{-10} Z \left(\frac{\alpha}{a_0^3}\right)^{1/2} \left(\frac{m_{\mathrm{H}}}{\mu}\right)^{1/2} \mathrm{cm}^{-3}\,\mathrm{s}^{-1} \tag{1.7}$$

Notably, it is independent of the temperature! So, in a cold gas, neutral-charged interactions have a higher collisional rate coefficient than the neutral-neutral scenario.

1.2 STATISTICAL MECHANICS

The interstellar medium (ISM) and intergalactic medium (IGM) are generally far from thermodynamic equilibrium. Nevertheless, the methods of statistical mechanics and thermodynamics provide powerful tools for understanding the non-equilibrium conditions that prevail, and for relating forward and reverse rates for the processes, such as ionization and recombination, that shape the medium.

But before getting any deeper than this, we should ask whether it is sensible of us to assume the ISM like a fluid. The general definition of fluid–which doesn't trouble the poor Knudsen and his number–is that of a macroscopic body with a bulk velocity. To achieve such a specific condition, the system we're trying to probe must have a negligible stochastic dispersion with respect to the bulk velocity on a certain scale. We'll look into those scales that satisfy $\lambda \gg l = 1/n\sigma$, where l is the mean free path of the particles gathered in the wannabe fluid. As a rule of thumb, you might want to use the following expression

$$l = \frac{1}{\sigma} = 55 \left(\frac{r}{1\,\text{Å}}\right)^{-2} \left(\frac{n}{1\,\mathrm{cm}^{-3}}\right)^{-1} \mathrm{AU} \tag{1.8}$$

1.2.1 Partition Function and Levels Population

Consider some physical system (e.g., atoms in a box of volume V) that is able to exchange energy with a "heat reservoir" at temperature T. The theory of statistical mechanics defines the partition function $\mathcal{Z}$ to be

Note that we're implicitly assuming the particles to be identical and the system to be isolated.

$$\mathcal{Z} = \sum_i g_i e^{-E_i/kT} \tag{1.9}$$

We know that the total number of particles N (which is fixed) must be equal to the sum of all the particles in the different energy states

$$N = \sum_i N_i \tag{1.10}$$

It is fairly easy, although tedius, to show that the number of particles in the i-th energy state is given by

$$N_i = \frac{N}{\mathcal{Z}(T)} g_i e^{-E_i/kT} \tag{1.11}$$

This configuration is crucial for the establishing of Local Thermodynamic Equilibrium: Matter obeys to the Boltzmann distribution for energy and Maxwell distribution for velocity, but LTE means that matter is not in *global* equilibrium with radiation interacting with it.

What if now we relax the condition of the particles being identical? The process grows immediately more tedious and verbose, but the procedure to follow is essentially the same.

If we get an isolated system with total energy E, total mass $\mathcal{M}$, with M reactant species, N product species: the most likely macroscopic state is that with maximum number of compatible microscopic states, that is the one that maximizes entropy. This time we need to find the ways in which the particles of each species can be combined.

Skipping all the calculations we find, instead of the number of particles, the volume densities that maximize the entropy

$$\boxed{\frac{\prod_{i=1}^{N} n_{P_i}^{b_i}}{\prod_{j=1}^{M} n_{R_j}^{b_j}} = \left[\frac{(2\pi kT)^{3/2}}{h^3}\right]^{\sum a_i - \sum b_j} \left[\frac{\prod_{i=1}^{N} M_{P_i^{b_i}}}{\prod_{j=1}^{M} M_{R_j^{b_j}}}\right]^{3/2} \frac{\prod_{i=1}^{N} z_{\text{int}}(P_i, T)^{b_i}}{\prod_{j=1}^{M} z_{\text{int}}(R_j, T)^{a_j}}} \tag{1.12}$$

which is the *law of mass action*. At equilibrium, the number of particles per unit volume ($n = N/V$) of each species involved is related only to temperature and to the partition functions of the individual species. The law is valid for every reaction or collision that happens in LTE.

Assume a reaction

$$R_1 + R_2 + ... + R_M \rightleftharpoons P_1 + P_2 + ... + P_N$$

so that the ratio between the collision rate coefficients is

$$\frac{\langle \sigma v \rangle_{\leftarrow}}{\langle \sigma v \rangle_{\rightarrow}} = \frac{\prod_{i=1}^{M} n_{R_i}}{\prod_{j=1}^{N} n_{P_j}}$$

Perhaps I'm tripping (fairly possible), but the law of detailed balance seems to be just the reciprocal of (1.12).

so that plugging in the law of mass action

$$\boxed{\left(\frac{\langle \sigma v \rangle_{\leftarrow}}{\langle \sigma v \rangle_{\rightarrow}}\right)^{-1} = \left[\frac{(2\pi kT)^{3/2}}{h^3}\right]^{N-M} \left[\frac{\prod_{i=1}^{N} M_{P_i^{b_i}}}{\prod_{j=1}^{M} M_{R_j^{b_j}}}\right]^{3/2} \frac{\prod_{i=1}^{N} z_{\text{int}}(P_i, T)^{b_i}}{\prod_{j=1}^{M} z_{\text{int}}(R_j, T)^{a_j}}} \tag{1.13}$$

results in the law of detailed balance.

This is fairly useful when applied to a realistic scenario, such as that of collisional (de)excitation

$$X(l) + Y \leftrightharpoons X(u) + Y$$

which has

$$\left(\frac{\langle \sigma v \rangle_{ul}}{\langle \sigma v \rangle_{lu}}\right) = \frac{z_{\text{int}}(X(l), T)}{z_{\text{int}}(X(u), T)} \implies \left(\frac{\langle \sigma v \rangle_{ul}}{\langle \sigma v \rangle_{lu}}\right) = \frac{g_l}{g_u} \exp\left(E_{ul}/kT\right) \tag{1.14}$$

The rate coefficients for a process of (de-)excitation of a given species are regulated by: (i) Temperature, (ii) degeneracy of involved levels, (iii) their energy difference.

Another example might be that of a generic chemical reaction

$$X + Y \leftrightharpoons Z + W$$

which has

$$\frac{\langle \sigma v \rangle_{\leftarrow}}{\langle \sigma v \rangle_{\rightarrow}} = \frac{z_{\text{int}}(X, T) z_{\text{int}}(Y, T)}{z_{\text{int}}(Z, T) z_{\text{int}}(W, T)} \propto \frac{\exp\left(-(E_X + E_Y)/kT\right)}{\exp\left(-(E_Z + E_W)/kT\right)} \tag{1.15}$$

1.3 ENERGY LEVELS OF MOLECULES

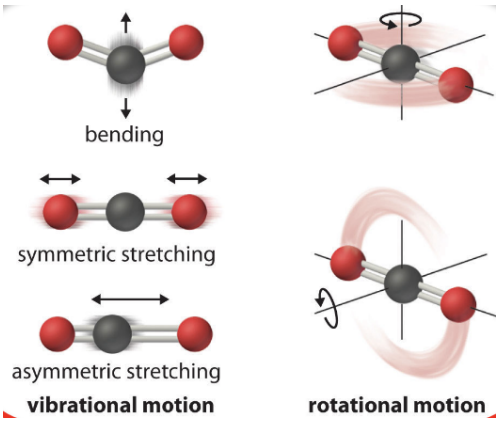
To properly discuss the formation and "position" of the energy levels of a given molecule, we should consider, in principle, quite a lot of contributions: The kinetic term, the attractive potentials of each atom, the electronic repulsion, and the electrostatic repulsion between the atoms themselves. The resulting Hamiltonian is rather complicated to treat as it is.

We then introduce the *Born-Oppenheimer approximation*, which assumes that Nuclei are essentially fixed and the electrons are free to move: The result is that the motion of the nuclei does not induce electronic transitions due to the different mass of electrons and nuclei. Under this approximation, you can demonstrate

$$E_{tot} = E_{el} + E_{vib} + E_{rot}$$

E_{el} *is the energy of electronic state,* E_{vib} *energy due to radial vibration of nuclei with respect to the equilibrium position,* E_{rot} *energy of rotation of nuclei about the centre of mass.*

As a further approximation, we'll assume that centrifugal distortion is negligible and only symmetric stretching is allowed.



Like in atoms, also in molecules the electronic transitions emit in the optical/UV bands, while the roto-vibrational transitions fall in the radio/mm/infrared bands. Note, however, that at the temperature of the molecular gas ($T < 100\,\mathrm{K}\ 10^{-2}\,\mathrm{eV}$) only the rotational levels in the ground vibrational and electronic states are populated!

The relevant parameters we'll need in our discussion are

- The classical energy E of a rigid rotator

$$E = \frac{J_a^2}{2I_a} + \frac{J_b^2}{2I_b} + \frac{J_c^2}{2I_c} \quad J^2 = J_a^2 + J_b^2 + J_c^2$$

  where a, b and c are the principal axis of the rotator;

- The permanent dipole moment μ. If the dipole moment is non-zero, electric dipole transitions are allowed. On the other hand, if the dipole moment is vanishing, only electric quadrupole transitions (and onwards) are allowed. These, however, are strongly disfavored.

In the following, we'll consider these configurations

- Diatomic, linear: $I_a = I_b \neq I_c = 0$;
- Symmetric-top: $I_a = I_b \neq I_c \neq 0$;
- Asymmetric-top: $I_a \neq I_b \neq I_c$;
- Spherical: $I_a = I_b = I_c$

Linear Rotors

The classical energy is easily expressed in terms of the total (square) angular momentum $E = J^2/2I$, which provides us a simple expression for the quantum version. Hence

$$E_j = B_{\mathrm{rot}} h j(j+1) \quad B_{\mathrm{rot}} = \frac{h}{8\pi^2 I}$$

and selection rules force transitions with $\Delta j = \pm 1$. Note that from (1.3) we deduce that $E_j \propto j^2$ and $h\nu = E_{j+1} - E_j \propto j$. Also note that $E_j \propto B_{\mathrm{rot}} \propto I^{-1}$, hence species with larger B_{rot} (at fixed j) have larger E_{j+1}, E_j and ν.

This has a rather troublesome implication: Molecular Hydrogen is only seen through absorption lines, and rarely from emission lines. This is for two reasons:

1. Molecular Hydrogen is *omonuclear*, so no dipole transitions but only electric quadrupole transitions $\Delta j = \pm 2$;
2. Molecular Hydrogen has *minimum moment of inertia*, so a huge separation between levels, and high T is needed to populate the excited states. Roto-vibrational transitions are visible only in very hot gas.

The lowest energy quadrupole transition for molecular Hydrogen has $\Delta E = 510\,\mathrm{K}$.

Symmetric-top Rotors

For the sake of simplicity, let us define $I_a = I_b = I_\perp$, and $I_c = I_\parallel$. The classical energy is thus written as

$$E = \frac{J^2}{2I_\perp} + J_c^2 \left(\frac{1}{2I_\parallel} - \frac{1}{2I_\perp} \right)$$

To the two quantum operators J^2 and J_c are associated the eigenvalues $j(j+1)\hbar^2$ and $m\hbar$, with $m \leq |j|$. We can define two rotational constants

$$A = \frac{h}{8\pi^2 I_\parallel} \quad B = \frac{h}{8\pi^2 I_\perp}$$

which we can use to write the rotational energy

$$E_{j,m} = Bhj(j+1) + (A-B)hm^2 \quad h\nu = 2hBj \tag{1.16}$$

and selection rules force transitions with $\Delta j = \pm 1$ and $\Delta m = 0$.

Asymmetric-top Rotors

This case is somewhat complicated and simple at the same time, because a general analytical expression that holds for all asymmetric-top rotators for E_j and ν_{ij} does not exist. The rotational energy will depend, not surprisingly, on two more quantum numbers, and we have to define three rotational constants.

Example: H_2O

The molecule of water has the two atoms of Hydrogen form an angle of 104.45°. It has a blatant axis of symmetry passing from the atom of Oxygen, thus meaning the molecule has three different momenta of inertia.
It is crucial to distinguish two possible configurations: *ortho* (the two Hydrogen atoms have parallel spins) and para water (the two Hydrogen atoms have anti-parallel spins). The complete wave function for the molecule of water (which is antisymmetric) is the product of four contributions $\psi = \psi_{el}\psi_{vib}\psi_{rot}\psi_s$, where the former two are symmetric, while the latter two must always have opposing parity.

- Ortho: m_{-1}, m_{+1} = even, odd (eo) or odd, even;
- Para: m_{-1}, m_{+1} = even, even (ee) or odd, odd;

but since selection rules force $\Delta j = 0, \pm 1$ and $\Delta m = \pm 1, \pm 3$, only transitions eo $\rightarrow$ oe or ee$\rightarrow$oo (and their inverse) are allowed, while transitions between ortho and para states are strongly forbidden.

Spherical Rotors

The expression this time is fairly simple

$$E = \frac{J^2}{2I} \implies E_j = \frac{j(j+1)\hbar^2}{2I} \quad h\nu = \frac{(j+1)\hbar^2}{I} \tag{1.17}$$

Note that non-polar molecules have no rotational dipolar lines unless they're distorted. Distortion, however, is only relevant for high values of j.

1.3.1 Lines Hyperfine Structure

Some nuclei have non-zero nuclear spin, which interacts (either via magnetic or electric quadrupole momentum) with electrons and/or the other nuclear spins. The energetic splitting due to this interaction gives the *line hyperfine structure*. Only some nuclei have non-zero nuclear spin, indicated usually with quantum number I, which depends on the particular combination of number of protons and neutrons in the nucleus. For example, the molecule $C^{17}O$ has the ^{17}O nucleus with nuclear spin $I = 5/2$ and presents the splitting shown in Fig.1.2.

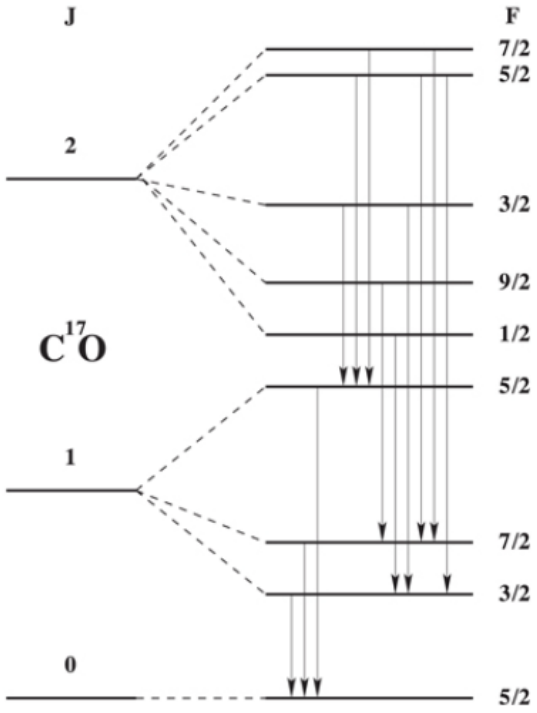


Figure 1.2: Hyperfine splitting induced by nuclear spin on the $C^{17}O$ molecule.

1.3.2 Inversion Transitions of NH_3

To address what's commonly known as inversion transition, as far as the molecule of Ammonia is concerned, we need mainly three elements:

1. *Motion*: Vibration of the N nucleus that goes up and down the plane of H nuclei;
2. *Associated potential*: The harmonic potential with a barrier centred on $z = 0$. The N nucleus can penetrate the barrier via quantum tunnelling;
3. *States*: The two quantum states at lowest energy associated with this potential, indicated as + and -, are the equivalent of those of the barrierless harmonic potential, and each rotational state defined by quantum

numbers (J, K) (NH_3 is symmetric-top) is splitted in the two inversion states + and -

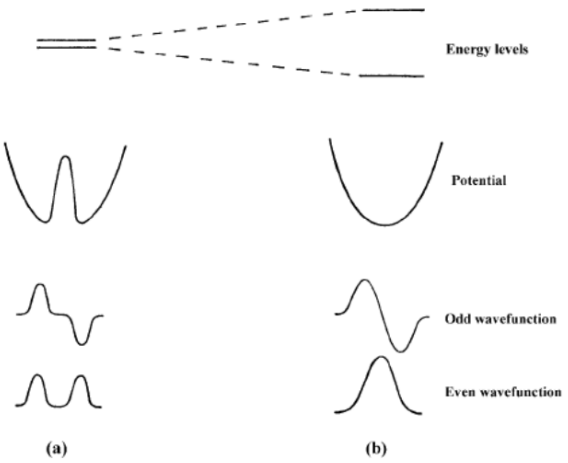


Figure 1.3: Development of energy level splitting when the potential barrier is reduced. (a) With barrier present, the wavefunctions have similar forms and energy levels are closely spaced; (b) as barrier is reduced, wavefunctions take on distinctly different form and energy levels separate.

The two inversion levels + and - are further split into hyperfine levels due to the non-zero spin of the N nucleus and of the three H nuclei. So, for example, in the rotational state $(j,k) = (1,1)$ the two inversion levels split into 5 hyperfine sub-levels; in total, the inversion transition (1,1) is splitted into 18 hyperfine components.

Vibrational Levels

In principle, we should also consider the energy level produced by vibration motions of biatomic molecules. Classically, we know that at large separations the Coulombian attraction of electrons on nuclei is dominating, while, on the shorter scales, prevails the Coulombian repulsion of nuclei. For biatomic molecules, the best approximation for this particular potential is the *Morse potential*

$$V(r) = D_e(1 - \exp(-a(r - r_e)))^2 \tag{1.18}$$

which we can Taylor-expand in a neighborhood of r_e, finding that the first non-vanishing order is the second one

$$V(r) = \frac{1}{2}k(r - r_e)^2 + o(r^3) \quad k = \left(\frac{\partial^2 V}{\partial r^2}\right)_{r=r_e} = 2a^2 D_e \tag{1.19}$$

and so the Morse potential is equivalent to an harmonic potential, which we know fairly well how to treat, also quantistically speaking. Note that unharmonic contributions grow (significantly) relevant moving to larger distances from the equilibrium position.

2 INTRODUCTION TO ASTROCHEMISTRY

Thanks to the advent and refining of radiotelescopes throughout the past century, it was possible to identify new molecules, both organic and inorganic, all around the ISM of the MW and, sporadically, in external galaxies (which is a considerable boon).

As of September 2025, $\sim$ 330 molecules were identified in the Milky Way and $\sim$ 74 in external galaxies. You can find an extensive list, if you're interested, at this link.

Not surprisingly, we've been detecting molecules of the most abundant and (chemically) active elements, that are Oxygen, Carbon, and Nitrogen.

2.1 SURFACE CHEMISTRY

The most elemental process is the radiative association of two atoms, A e B to form the molecule AB. The molecule will have an internal energy

$$A + B \rightarrow AB^*$$

The two associating atoms form a compound known as activated complex (AB^*) which will usually radiate away the excess energy produced by its formation. AB^* must lose this excess energy during the impact to remain bound and not dissociate immediately. This energy loss through emission of photons is a *very rare process*. However, the molecule can lose this excess energy via three-body collisions

$$A + B \rightarrow AB^* + M \rightarrow AB + M^*$$

which are still rather rare in the ISM, where density is at most $\sim 10^7 - 10^{10}\,\mathrm{cm}^{-3}$. De-excitation through three-body collisions is much more probable in denser environments like Earth's atmosphere.

So how can we prevent the radiative dissociation of molecules in the ISM? Dust grains, thanks to their very large surface and heat capacity, can favor atom encounters and easily absorb the ecess energy.

The process goes through three fundamental steps, which are also pictorially shown in Fig.2.1:

1. *Accretion*: the particle impacts the grain surface and can be *adsorbed*[1] if the temperature is lower than the evaporation temperature. Accretion can occur via two dominant mechanisms
   - *Physisorbption*: van der Waals forces ($0.01 - 0.2\,\mathrm{eV}$);
   - *Chemisorbption*: chemical bond ($1\,\mathrm{eV}$).

1 i.e. can stick on surface. And no, it's not a typo.

2. *Surface diffusion*: if the particle is not chemisorbed, it can move on the grain surface, meet other species and, maybe, react to form new molecules.

3. *Desorption*: if temperature increases above the evaporation temperature, the new molecule goes in the gas.

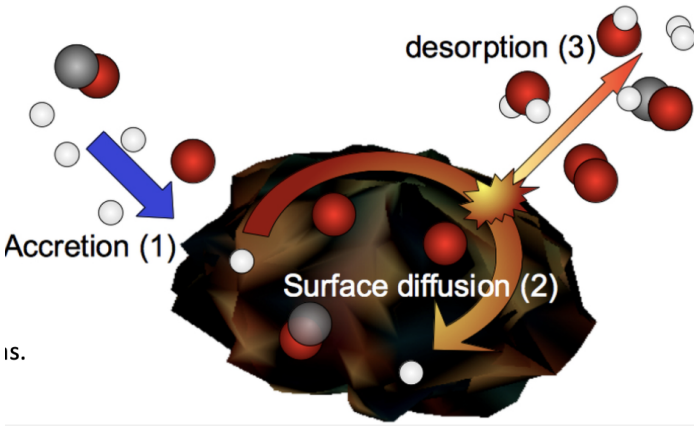


Figure 2.1: The three fundamental steps of surface chemistry on dust grains.

The "sites" where the probability to form new molecules is highest are the gravitational potential walls on the (non-smooth) dust grains. Transition from one site to the other can happen via two mechanisms: Thermal hopping[2] (the adsorbed species migrate from one site to the other exploting the warming of the dust grain), and the good old quantum tunneling.

Hydrogen (and Deuterium) is the most abundant and most "mobile" reactive species. Therefore, hydrogenation (and deuteration) is a very efficient surface chemistry process

$$
\begin{aligned}
\mathrm{H} &\rightarrow \mathrm{H_2} \\
\mathrm{O} &\rightarrow \mathrm{OH} \rightarrow \mathrm{H_2O} \\
\mathrm{C} &\rightarrow \mathrm{CH} \rightarrow \mathrm{CH_2} \rightarrow \mathrm{CH_3} \rightarrow \mathrm{CH_4} \\
\mathrm{N} &\rightarrow \mathrm{NH} \rightarrow \mathrm{NH_2} \rightarrow \mathrm{NH_3} \\
\mathrm{CO} &\rightarrow \mathrm{HCO} \rightarrow \mathrm{H_2CO} \rightarrow \mathrm{H_3CO} \rightarrow \mathrm{CH_3OH}
\end{aligned}
$$

If you're interested in the subject, you might want to take a look at [1], [4], [12], [15], and many others for which I didn't manage to pinpoint the exact article.

2.2 GAS-PHASE CHEMISTRY

After H_2 was formed and evaporated from grains, the gas-phase chemistry starts. H_2 is so much more abundant than any other molecule that if a species can react with H_2, no other reactions need to be considered. How

2 You can read, for example, [10].

can we understand if a reaction with H_2 is spontaneous? A reaction at p and T constant is spontaneous if the Gibbs' free energy

$$\Delta G = \Delta H - T\Delta S = \Delta U + RT\Delta N - T\Delta S < 0 \tag{2.1}$$

In particular, a reaction is spontaneous at low temperature if $\Delta U < 0$, hence only molecular binding energy is involved.

Note that this condition holds intil the gas is isolated. Clearly, energy injection from external factors might provide the energy needed by endothermic reactions to start.

Consider for example the reaction

$$C + H_2 \rightarrow CH + H$$

Molecular Hydrogen has a binding energy of 4.48 eV, while CH has 3.47 eV. Hence, the reaction is endothermic $\Delta U = U_r - U_p > 0$ and cannot occur spontaneously in molecular clouds, where typically $kT < 0.01$ eV.

In the molecular ISM, only exothermic reactions can occur!

2.2.1 Neutral-Neutral Reactions

Consider a reaction like

$$A + BC \rightarrow AB + C$$

Reactions of this sort present an *activation energy*, which is the minimum energy to break the existing bond. Usually, this energy is of the order of $E_a \sim 0.11 - 1$ eV while the temperature of the ISM is $T \lesssim 0.01$ eV. This means that only exothermic and barrierless reactions can occur in the molecular ISM.

2.2.2 Neutral-Charged Reactions

Also called Ion-Neutral, they involve interactions of the sort of

$$A^+ + BC \rightarrow AB^+ + C$$

Ion-neutral exothermic reactions typically do not have activation barrier, thanks to the long range attractive force between the ion and the induced dipole moment (see Chapter 1, or [2] and [9]). This is because the collision rate coefficient has no dependence from the temperature of the region for this particular type of interaction.

2.3 FORMATION OF H_3^+

After the formation of molecular Hydrogen. cosmic rays can ionize the molecule

$$H_2 + \text{c.r.} \rightarrow H_2^+ + e^- + \text{c.r.}$$

Once H_2^+ is formed in small amounts, it reacts very quickly with the very abundant H_2 to form H_3^+ the most abundant and important molecular ion in the ISM

$$H_2 + H_2^+ \rightarrow H_3^+ + H$$

This new ion triggers ion-neutral reactions that lead to the formation of more complex molecules. It can an easily donate a proton to some neutrals via exothermic reactions, e.g. with atomic O, and produce another ion. If this ion can react with H_2, it gives rise to another ion, and so on. The chain of reactions with H2 stops when the final ion cannot react with H_2 anymore for energetic reasons.

For example

$$\begin{aligned} H_3^+ + O &\rightarrow OH^+ + H_2 \\ OH^+ + H_2 &\rightarrow H_2O^+ + H \\ H_2O^+ + H_2 &\rightarrow H_3O^+ + H \\ H_3O^+ + e^- &\rightarrow H_2O + H \\ &\rightarrow OH + H_2 \implies OH + O \rightarrow O_2 + H \\ &\rightarrow O \end{aligned}$$

It can also prompt the formation of simple hydrocarbons: The formation of carbon-bearing species starts with a sequence of reactions very similar to those that initiate the chemistry of oxygen-bearing species. It procedes through a series of *proton transfers* from H_3^+ to a neutral atom, forming a hydrogenated ion, then to *Hydrogen abstraction reactions*, which is a chain of reactions in which one H of H_2 is transferred to an ion. It stops to an ion that cannot react with H_2 anymore, and finally ends with *dissociative recombination*, the ion that cannot react with H_2 dissociates upon recombination with an electron.

Then again, H_3^+ also prompts the initiation of Nitrogen chemistry which, however, starts from

$$CH + N \rightarrow CN + H$$

2.4 FORMATION AND DESTRUCTION OF CO

Carbon Monoxyde is formed thanks to a series of chemical reactions

$$\begin{aligned} C + H_3O^+ &\rightarrow HCO^+ + H_2 \\ O + CH_3 &\rightarrow HCO^+ + H_2 \\ HCO^+ + e^- &\rightarrow CO + H \end{aligned}$$

Carbon Monoxyde is very stable ($B \sim 11.09\,\text{eV}$) and difficult to destroy. Curiously enough, it is destroyed by the very ion that prompts the start of all the fancy chemistry we've described in the last section

$$H_3^+ + CO \rightarrow HCO^+ + H_2$$

but the last reaction of the previous cycle immediately reforms CO.

More efficient ways to "remove" Carbon Monoxyde are

$$\begin{aligned} He^+ + CO &\rightarrow He + C^+ + O \\ h\nu + CO &\rightarrow C + O \end{aligned}$$

but part of C^+ reacts with OH and water (but not with molecular Hydrogen, since the reaction is endothermic)

$$\begin{aligned} C^+ + OH &\rightarrow CO^+ + H \\ CO^+ + H_2 &\rightarrow HCO^+ + H \\ C^+ + H_2O &\rightarrow HCO_3^+ + H \end{aligned}$$

while most photons able to dissociate CO (so with $h\nu > 11.09\,\mathrm{eV}$) are absorbed in advance from H_2.

2.5 FREEZE-OUT IN COLLAPSING CLOUDS

Let us consider a collapsing cloud: as long as the depletion timescale is longer than the free-fall timescale, the *freeze-out*[3] does not occur during collapse. But if τ_{dep} is shorter than τ_{ff}, the molecule freezes during collapse, disappearing from the gas! In the solid phase, it cannot be "seen" in emission anymore. In short, we have to compare

$$\tau_{ff} = \left(\frac{3\pi}{32G\rho}\right)^{-1/2} = 4 \cdot 10^{-7} n_{\mathrm{H}}^{-1/2}\,\mathrm{yr} \tag{2.2}$$

with

$$\tau_{\mathrm{dep}} = \frac{1}{\alpha n_d \sigma_d v_t} \approx 10^9 \sqrt{\frac{m_X}{T}} (n_{\mathrm{H}}\alpha)^{-1}\,\mathrm{yr} \tag{2.3}$$

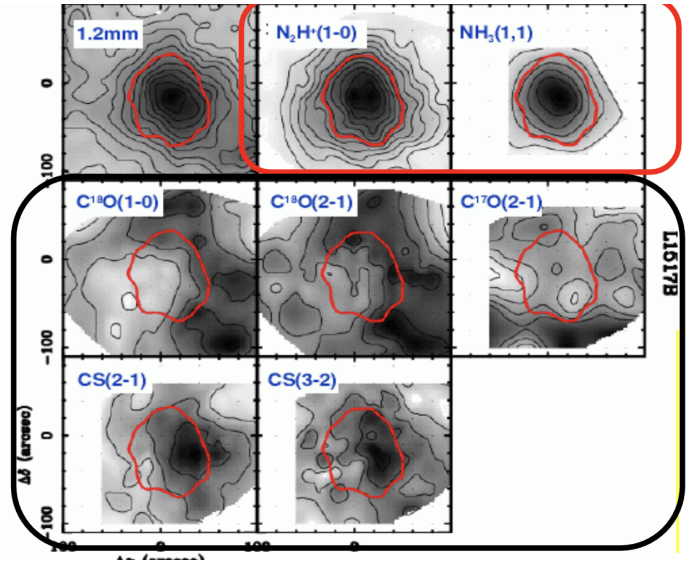


Figure 2.2: How a collapsing core looks like in the emission of gas and dust.

3 Please, do not mistake this freeze-out with other kinds of freeze-outs, like neutrons freeze-out and the sorts. Here the freezing is quite literal, while in the other case is more of a figure of speech to hint at the fact that their abundance is no longer changing.

2.6 ISOTOPIC FRACTIONATION

We define *isotopic fractionation* the the chemical process(es) by which the fraction of heavier isotopes in molecules varies through chemical reactions.

For example

$$^{12}\mathrm{CO} + ^{13}\mathrm{C}^{+} \leftrightharpoons ^{12}\mathrm{C}^{+} + ^{13}\mathrm{CO} + 35\,\mathrm{K}$$

The difference in molecular binding energy between reactants and products, caused by different zero-point vibration energy, makes one direction exothermic, i.e. favoured, in cold gas: The molecule with heavier isotope has lower *zero-point energy*.

The zero-point energy of a biatomic molecule is usually expressed through the equation

$$E_\nu \approx \frac{1}{2} g \left(\frac{1}{2\pi} \sqrt{\frac{k}{m_r}} \right) \tag{2.4}$$

If temperature is lower than $T \leq 20\,\mathrm{K}$, then

$$\mathrm{H}_3^+ + \mathrm{HD} \leftrightharpoons \mathrm{H}_2\mathrm{D}^+ + \mathrm{H}_2 + 230\,\mathrm{K}$$

and the ratio $\mathrm{H}_2\mathrm{D}/\mathrm{H}_3^+$ increases of orders of magnitude since

$$\frac{n(\mathrm{H}_3^+)}{n(\mathrm{H}_2\mathrm{D}^+)} \frac{n(\mathrm{HD})}{n(\mathrm{H}_2)} = \frac{\langle \sigma v \rangle_{\leftarrow}}{\langle \sigma v \rangle_{\rightarrow}}$$

which implies that

$$\frac{n(\mathrm{H}_3^+)}{n(\mathrm{H}_2\mathrm{D}^+)} = \frac{n(\mathrm{H}_2)}{n(\mathrm{HD})} \exp\left(-\frac{\Delta E}{kT} \right)$$

On the other hand, if CO depletes ($n \gtrsim 10^5\,\mathrm{cm}^{-3}$), H_3^+ and $\mathrm{H}_2\mathrm{D}^+$ remain abundant

$$\mathrm{H}_3^+ + \mathrm{X} \leftrightharpoons \mathrm{XH}^+ + \mathrm{H}_2$$
$$\mathrm{H}_2\mathrm{D}^+ + \mathrm{X} \leftrightharpoons \mathrm{XD}^+ + \mathrm{H}_2$$

Both the conditions we presented are typical in dense and cold cores: The ratio $\mathrm{XD}^+/\mathrm{XH}^+$ is orders of magnitude larger than the [D/H] ISM value (10^{-5}).

3 RADIATIVE TRANSFER

3.1 INTRODUCTION

Most of our knowledge about the Universe is based on the electromagnetic radiation that reaches us from far far away. EM radiation is obviously not the only way we can probe the Universe we live in but, in respect to neutrinos, cosmic rays or even gravitational waves, it's not a long stretch to claim it is by far the most understood.

It is most important then that an astrophysicist worthy of his (or her) name has a good grasp of the theory of radiative transfer and of its applications.

Apart from a few more key differences, I'll follow the description of radiative transfer of [3] and [13], but I won't fail to emphasize whenever I'll be doing otherwise.

3.2 RELEVANT QUANTITIES FOR RADIATIVE TRANSFER

Although some books often start their description of radiative transfer from the definition of *monochromatic energy* and *monochromatic intensity*, I found that it is most misleading, since, in all but a few cases, what we experimentally measure are fundamentally *fluxes*.

We shall then consider the *monochromatic flux* F_ν ($\mathrm{erg\,s^{-1}\,Hz^{-1}\,cm^{-2}}$) produced by some source passing through a small area $\mathrm{d}A$ located somewhere in space.

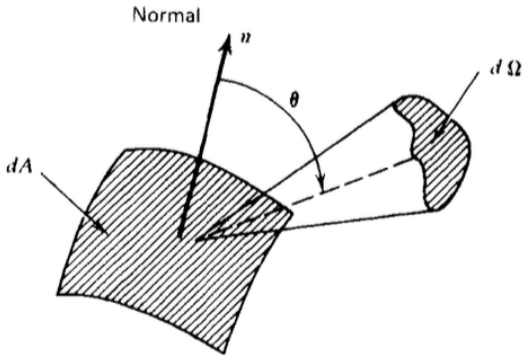


Figure 3.1: Schematic geometrical representation of the system. Credits: [13].

If we call $\hat{k}$ the propagation direction of the flux and $\hat{n}$ the unit vector emerging from the surface $\mathrm{d}A$, it's easy to get convinced that what is actually passing through the surface is something proportional to $F_\nu(\hat{k}\cdot\hat{n})$.

From the monochromatic flux we can define the *bolometric flux*, which is just the monochromatic flux integrated over all frequencies (or wavelengths)

$$F = \int_0^{+\infty} F_\nu \, \mathrm{d}\nu = \int_0^{+\infty} F_\lambda \, \mathrm{d}\lambda \tag{3.1}$$

This also tells us how to convert a flux per unit frequency to a flux per unit wavelength

$$F_\nu \, \mathrm{d}\nu = F_\lambda \, \mathrm{d}\lambda$$

By now it should be clear that, despite being experimentally sensible to use the flux, we're losing much information sticking with it, namely directional information.

We consider then the amount of radiation $E_\nu \, \mathrm{d}\nu$ passing through the same area in time $\mathrm{d}t$ and solid angle $\mathrm{d}\Omega$. Hence we can write

$$\mathrm{d}E_\nu \, \mathrm{d}\nu = I_\nu(\mathbf{r}, t, \hat{k})(\hat{k} \cdot \hat{n}) \, \mathrm{d}t \, \mathrm{d}\Omega \, \mathrm{d}A \, \mathrm{d}\nu \tag{3.2}$$

where the quantity $I_\nu(\mathbf{r}, t, \hat{k})$ is called the *specific monochromatic intensity*. If $I_\nu(\mathbf{r}, t, \hat{k})$ is specified for all directions at every point in a certain region of spacetime, then we'd have a complete prescription of the radiation field we intend on studying.

Capitalizing on the blatant similarities with distribution functions, we can evaluate the moments of the monochromatic intensity.

Definition 3.2.1. *Monochromatic mean intensity J_ν*

$$J_\nu = \frac{1}{4\pi} \int_\Omega I_\nu \, d\Omega = \frac{c}{4\pi} U_\nu$$

with U_ν the total energy density of radiation. Note that J_ν is essentially just an average of the monochromatic intensity over all solid angles.

Definition 3.2.2. *Monochromatic flux $\vec{F}_\nu$*

$$\vec{H}_\nu = \frac{1}{4\pi} \int_\Omega I_\nu(\hat{k}) \hat{k} \, d\Omega \implies \frac{1}{4\pi} F_\nu = \vec{H}_\nu \cdot \hat{n}$$

I haven't explicitly proved the last equality, but it shouldn't be hard for you to convince yourself (or prove it yourself) that it is indeed true.

Definition 3.2.3. *Monochromatic radiation pressure p_ν The monochromatic pressure is defined starting from the different directions correlations of the monochromatic intensity*

$$K_\nu^{ij} = \frac{1}{4\pi} \int_\Omega I_\nu(\hat{k}) n^i n^j \, d\Omega$$

The pressure in particular is usually expressed as

$$P_\nu = \frac{1}{c} \int_\Omega I_\nu(\hat{k}) \cos^2\theta \, d\Omega$$

where $\cos^2\theta = (\hat{k} \cdot \hat{n})^2$.

In particular, note that for an isotropic field I_ν, the average brightness J_ν is one and the same with the monochromatic specific intensity

$$J_\nu = I_\nu \quad \text{(if the field is isotropic)}$$

3.3 BLACKBODY FUNCTION

Even at an undergraduate level, we're all fairly familiar with *blackbody radiation*. The easiest way to deduce the expression for the energy density of photons in *thermal equilibrium* (STE) inside a cavity is by the means of statistical mechanics.

Remember the Bose-Einstein distribution ($\mu = 0$)

$$n = \frac{1}{\exp(h\nu/kT) - 1}$$

and the phase space density of states (per unit volume)

$$\rho(\nu)\,\mathrm{d}\nu = \frac{4\pi h g \nu^3}{c^3}\,\mathrm{d}\nu$$

from which deducing the expression from internal energy is straightforward. Remembering $g = 2$ is the quantum degeneracy of photons, a simple moltiplication of the previous expressions yields

$$U_\nu\,\mathrm{d}\nu = \frac{8\pi h}{c^3}\frac{\nu^3}{\exp(h\nu/kT) - 1}\,\mathrm{d}\nu \tag{3.3}$$

The energy of a photon is $E_\gamma = h\nu$, meaning the number of photons in the frequency range $\nu \to \nu + \mathrm{d}\nu$ is

$$n_\nu\,\mathrm{d}\nu = \frac{8\pi}{c^3}\frac{\nu^2}{\exp(h\nu/kT) - 1}\,\mathrm{d}\nu \tag{3.4}$$

which can be integrated over all frequencies, thus obtaining the number density of photons in blackbody radiation

$$n_\gamma = \beta T^3 \quad \beta = \frac{2.4041 k^3}{\pi^2 \hbar^3 c^3} \tag{3.5}$$

Since blackbody radiation is isotropic (it depends only on the absolute temperature T), the definition of mean monochromatic intensity yields

$$\boxed{B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{\exp(h\nu/kT) - 1}} \tag{3.6}$$

It's important to notice that, in principle, such a fundamental result holds only in *strict thermodynamic equilibrium* (STE), but it's possible to generalize this formulation for less "restrictive" environments.

An incredible number of important results descends from (3.6), and it may be worthwhile to cite at least some of them, starting from Stefan-Boltzmann law. We'll use the following result without proving it

$$\int_0^{+\infty} B_\nu(T)\,\mathrm{d}\nu = \frac{2h}{c^2}\frac{\pi^4}{15}\left(\frac{kT}{h}\right)^4$$

Computing the bolometric flux and the bolometric energy density by integrating over all frequencies using what we've just written down, you find the following

$$U(T) = aT^4 \quad F(T) = \sigma_{SB} T^4$$

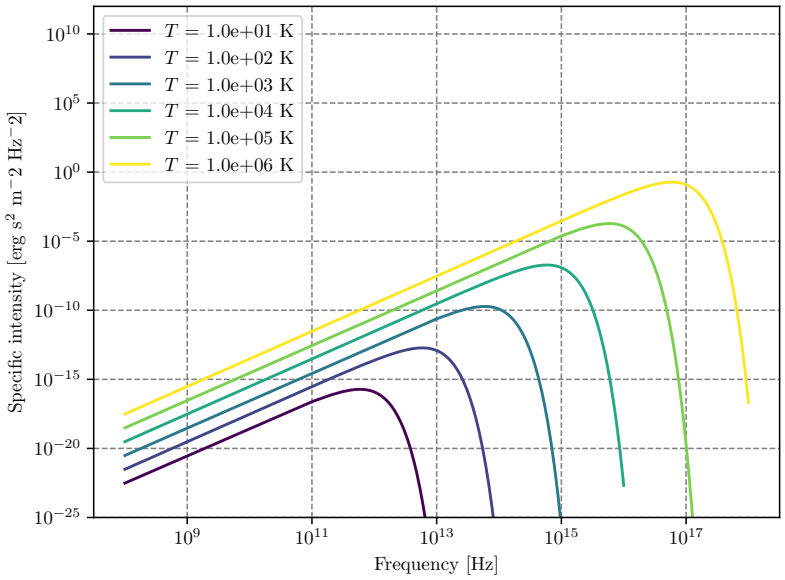


Figure 3.2: Blackbody frequency spectrum.

Clearly the two constants a and σ_{SB} cannot be independent, and are actually related by the integral we've previously calculated. Using for example[1]

$$F(T) = \pi \int_0^{+\infty} B_\nu(T)\, \mathrm{d}\nu$$

you can easily find out that the *Stefan-Boltzmann constant* is equal to

$$\sigma_{SB} = \frac{2\pi^5 k^4}{15c^2h^3}$$

and the relation with a is simply $\sigma_{SB} = ac/4$.

The equation

$$F(T) = \frac{2\pi^5 k^4}{15c^2h^3} T^4 \tag{3.7}$$

is what is usually known as the *Stefan-Boltzmann law*.

Let us now consider two different regimes for eq.3.6: $h\nu/kT \ll 1$ and $h\nu/kT \gg 1$. The first yields what is commonly known as the Rayleigh-Jeans Law which is, sadly, pretty much relevant only for radioastronomy.

Since

$$\exp\left(\frac{h\nu}{kT}\right) = 1 + \frac{h\nu}{kT} + o\left(\frac{h\nu}{kT}\right)^2$$

the blackbody radiation assumes the much simpler form of

$$B_\nu^{RJ} = \frac{2\nu^2}{c^2} kT$$

Another important results is achieved in the opposite regime, when the exponential term is rather larger than unity

$$B_\nu^{W} = \frac{2h\nu^3}{c^2} \exp\left(-\frac{h\nu}{kT}\right)$$

1 The emergent flux from an isotropically emitting surface (such as a blackbody) is $\pi \cdot$ brightness which is none other than the specific intensity.

This expression is known as Wien's Law.

Characteristic Temperatures

Starting from the blackbody spectrum we can give various definitions of temperature, that are going to be more and less useful when dealing with stars. The definitions are those of [13].

Brightness Temperature

One way of characterizing brightness (specific intensity) at a certain frequency is to give the temperature of the blackbody having the same brightness at that frequency. That is, for any value I_ν we define $T_b(\nu)$ by the relation

$$I_\nu = B_\nu(T_b)$$

this way of specifying brightness has the advantage of being closely connected with the physical properties of the emitter, and has the simple unit (K). This procedure is used especially in radio astronomy, where the Rayleigh-Jeans law is usually applicable.

$$T_b = \frac{c^2}{2\nu k} I_\nu \quad h\nu/kT \ll 1 \tag{3.8}$$

Note that the uniqueness of the definition of brightness temperature relies on the monotonicity property of Planck's law. Also note that, in general, the brightness temperature is a function of ν.

Color Temperature

Often a spectrum is measured to have a shape more or less of blackbody form, but not necessarily of the proper absolute value. For example, by measuring F_ν, from an unresolved source we cannot find I_ν, unless we know the distance to the source and its physical size.

By fitting the data to a blackbody curve without regard to vertical scale, a color temperature T_c, is obtained. Often the "fitting" procedure is nothing more than estimating the peak of the spectrum and applying Wien's displacement law to find a temperature.

Effective Temperature

The effective temperature of a source T_{eff} is derived from the total amount of flux, integrated over all frequencies, radiated at the source. We obtain T_{eff} by equating the actual flux F to the flux of a blackbody at temperature T_{eff}

$$F = \int I_\nu \cos\theta \, \mathrm{d}\nu \, \mathrm{d}\Omega := \sigma T_{eff}^4 \tag{3.9}$$

3.4 RADIATIVE TRANSFER EQUATION

In the presence of matter, it is not immediately obvious what changes may occur in the specific intensity as we move along a ray path. The aim of this section will be to eviscerate the matter.

Let's consider the following geometric construction

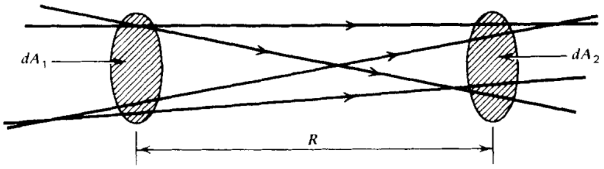


Figure 3.3: Geometrical construction for ray paths propagating in empty space. Credits: [13].

It won't take a lot of effort to convince yourself that in empty space the monochromatic intensity I_ν is actually conserved. In fact, from simply writing down the definitions and imposing the conservation of energy

$$I_{\nu_2}\,\mathrm{d}A_2\,\mathrm{d}t\,\mathrm{d}\Omega_2\,\mathrm{d}\nu = I_{\nu_1}\,\mathrm{d}A_1\,\mathrm{d}t\,\mathrm{d}\Omega_1\,\mathrm{d}\nu$$

the conclusion follows observing that $\mathrm{d}A_2\,\mathrm{d}\Omega_2 = \mathrm{d}A_1\,\mathrm{d}\Omega_1$.

If we consider an affine parameter of the form $\vec{x} = \vec{x}_0 + \hat{k}s$, we may as well write the previous results in a more familiar fashion

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = 0 \implies (\hat{k}\cdot\nabla)I_\nu = 0 \tag{3.10}$$

What changes if matter is present along the ray path? Clearly it will no longer be true that $(\hat{k}\cdot\nabla)I_\nu = 0$, but we're not that far off. All that we need is some little work on both terms.

How the right hand side of the equation should change is obvious: It needs to keep track of the "creation" and "destruction" of photons in the considered volume of spacetime.

The left hand side of the equation requires a little more care. Consider infinitesimal time and space displacements along the ray path, respectively $\mathrm{d}t$ and $\mathrm{d}\vec{x}$

$$\Delta E_\nu\,\mathrm{d}\nu = \Big(I_\nu(\vec{x}+\mathrm{d}\vec{x}, t+\mathrm{d}t, \hat{k}) - I_\nu(\vec{x}, t, \hat{k})\Big)\,\mathrm{d}t\,\mathrm{d}\Omega\,\mathrm{d}A\,\mathrm{d}\nu$$

Taking a first order expansion in respect to the affine parameter s along the ray path yields

$$\left(\frac{1}{c}\partial_t I_\nu + \partial_s I_\nu\right)\,\mathrm{d}t\,\mathrm{d}s\,\mathrm{d}\Omega\,\mathrm{d}A\,\mathrm{d}\nu = \text{photon addition} - \text{photon removal}$$

This equation is a generalization of eq.3.10 for non-stationary radiative transport and in the presence of matter. It's about time we get to know what "lives" in the right hand side of the equation.

3.4.1 Monochromatic emission coefficient

For the moment, we'll define the *spontaneous* monochromatic emission coefficient j_ν as

$$\mathrm{d}E_\nu \, \mathrm{d}\nu = j_\nu \, \mathrm{d}V \, \mathrm{d}t \, \mathrm{d}\Omega \, \mathrm{d}\nu \tag{3.11}$$

which in general has a non-zero dependence on the emission direction. Sometimes the spontaneous emission coefficient is defined by the *emissivity*[2] ϵ_ν, which is the energy emitted spontaneously per unit frequency per unit time per unit mass

$$j_\nu = \frac{\epsilon_\nu \rho}{4\pi}$$

where ρ is the mass density of the emitting medium.

If we perform the decomposition $\mathrm{d}V = \mathrm{d}A \, \mathrm{d}s$, the contribution of spontaneous emission to the specific intensity is

$$\mathrm{d}I_\nu = j_\nu \, \mathrm{d}s$$

3.4.2 Absorption coefficient

Similarly, we can consider the energy that is absorbed from the radiation when passing through a medium. There exists various definitions; I'll use the one we gave in class and that is incidentally the one used in [3] and [13] as well.

We define the *absorption coefficient* α_ν through the following expression

$$\mathrm{d}I_\nu = -\alpha_\nu I_\nu \, \mathrm{d}s \tag{3.12}$$

If we use a microscopic model, then the absorption coefficient can be understood as particles with numeric density n presenting an effective absorbing area, the *cross section* σ_ν. The coefficient α_ν can thus be rewritten in terms of

$$\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$$

where κ_ν is called the mass absorption coefficient or the *mass-weighed opacity coefficient*.

Note that in eq.3.12, we consider "absorption" to include both "true absorption" and stimulated emission, because both are proportional to the intensity of the incoming beam. Depending on the entity of the contribution, the α_ν coefficient may be positive or even negative, giving raise to curious phenomena.

Making full use of what we've just defined, we can finally present the celebrated *equation of radiative transfer* (although in the notable absence of scattering)

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = -\alpha_\nu I_\nu + j_\nu \tag{3.13}$$

which is actually fairly easy to solve when one of the two coefficients vanishes.

2 Please note that often the two names are used almost interchangeably.

Emission only

We set $\alpha_\nu = 0$ and the equation may be solved by direct integration

$$I_\nu(s) = I_\nu(s_0) + \int_{s_0}^{s} j_\nu(s')\,\mathrm{d}s'$$

the result is not that interesting per se.

Absorption only

This time we set $j_\nu = 0$. The equation is easily solved this time as well

$$I_\nu(s) = I_\nu(s_0)\exp\left(-\int_{s_0}^{s}\alpha_\nu(s')\,\mathrm{d}s'\right)$$

In this case, it's rather common to write down the equation in terms of a new variable, the *optical depth*[3] τ_ν

$$\mathrm{d}\tau_\nu = -\alpha_\nu\,\mathrm{d}s \tag{3.14}$$

which means

$$\tau_\nu = -\int_{s_{\text{out}}}^{s_{\text{in}}}\alpha_\nu(s)\,\mathrm{d}s \tag{3.15}$$

Given this definition we'll say that if

- $\tau_\nu \gg 1$: the medium is *optically thick or opaque*
- $\tau_\nu \ll 1$: the medium is *optically thin or transparent*

This has some crucial implications we'll be going through in a moment.

In the stationary limit, the equation of radiative transport may be written as

$$(\hat{k}\cdot\nabla)I_\nu(\hat{k},\vec{x}) = j_\nu(\vec{x}) - \alpha_\nu(\vec{x})I_\nu(\hat{k},\vec{x})$$

In terms of the *source function* $S_\nu = j_\nu/\alpha_\nu$ it becomes (dropping the functional dependencies for conciseness)

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}\tau_\nu} = -I_\nu + S_\nu \tag{3.16}$$

which can be integrated to yield the formal solution

$$\boxed{I_\nu(\hat{k},\tau_\nu) = I_\nu(\tau_{\nu,0})\exp(-\tau_\nu) + \int_{\tau_{\nu,0}}^{\tau_\nu} d\tau'_\nu\, S_\nu \exp(-(\tau_\nu - \tau'_\nu))} \tag{3.17}$$

Assume for the moment that the matter through which radiation is passing has constant properties and has no background source. Then the source function S_ν is constant and the formal equation becomes

$$I_\nu = S_\nu(1 - e^{-\tau_\nu})$$

If the medium and is optically thin, then the equation is reduced to

$$I_\nu = S_\nu\tau_\nu = j_\nu L \tag{3.18}$$

3 Although I don't support this particular choice–since it is bound to create confusion with the mass-weighed opacity κ_ν–the optical depth is sometimes referred to as simply *opacity*.

by taking the Taylor expansion of the exponential term and calling L some typical length of the medium.

If, on the other hand, the medium is optically thick, we can neglect the exponential $e^{-\tau_\nu}$ to obtain

$$I_\nu = S_\nu \tag{3.19}$$

3.5 KIRCHHOFF'S LAW AND LTE

The most notable implication of (3.19) is if we consider the specific intensity coming out of a small hole on a box kept in thermodynamic equilibrium. We know that what's going to come out of there is the blackbody radiation

$$I_\nu = B_\nu(T)$$

but what if we were to put an optically thick object just behind the hole?

If the object is in thermodynamic equilibrium with the surroundings (and it *will* be, given an appropriate amount of time), then the radiation coming out of the hole will still be blackbody radiation. But (3.19) tells us that the source function will tend to be equal to the specific intensity, hence

$$S_\nu = B_\nu(T) \tag{3.20}$$

which actually puts a constraint on the possible values of the emission coefficient in terms of the absorption coefficient. This is exactly what is expressed in Kirchhoff's law

$$\boxed{j_\nu = \alpha_\nu B_\nu} \tag{3.21}$$

Let us briefly consider what we have just derived. Matter often tends to emit and absorb at specific frequencies corresponding to what are commonly called *spectral lines*. We would expect then both j_ν and α_ν to have peaks (or depressions) around these lines. But Kirchhoff's law forces their ratio to be equal to a smooth blackbody profile.

Thus we can expect to observe two very different scenarios if the medium is optically thin rather than optically thick. In the former, the radiation emerging from the medium is essentially determined by its emission coefficient; since j_ν is expected to present peaks, so will the radiation spectrum, which will be appear in spectral lines, as shown in Fig.3.4 and Fig.3.5.

On the other hand, the intensity coming out of an optically thick body is its source function, which must be equal to the blackbody function. Hence we expect the medium to emit in a continuum, like a blackbody.

All throughout this description, we've been assuming the medium to have (global) constant properties, which has the perk of being a good approximation for many objects of interest, but still turns out to be a really poor one for many other objects. Stars, for example.

Ingenuously, we may expect stars to emit radiation like blackbodies, but they're not. Actually, stars present absorption lines, many, even, depending on the class of star. What we cannot assume in stars is them having constant properties, starting from temperature.

In fact, we could take a guess and claim that stars are in *strict* thermodynamic equilibrium. It would be a very bad guess indeed.

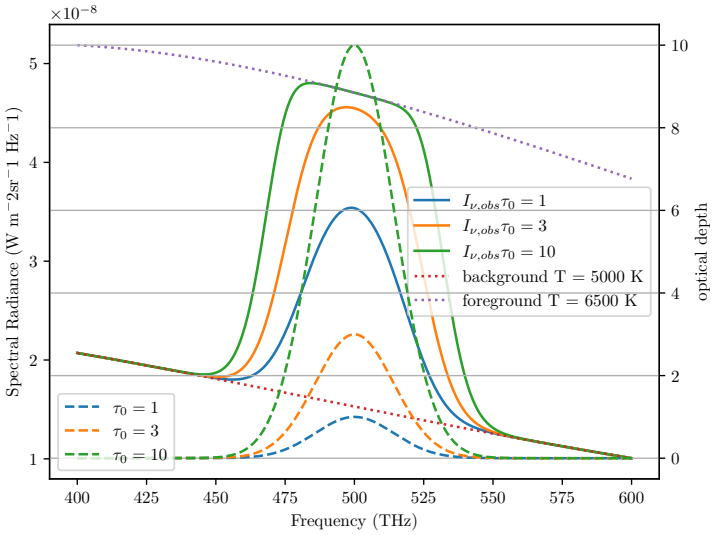


Figure 3.4: An example of emission features formation for different temperatures and different values of τ. Credits: Prof. Walter del Pozzo.

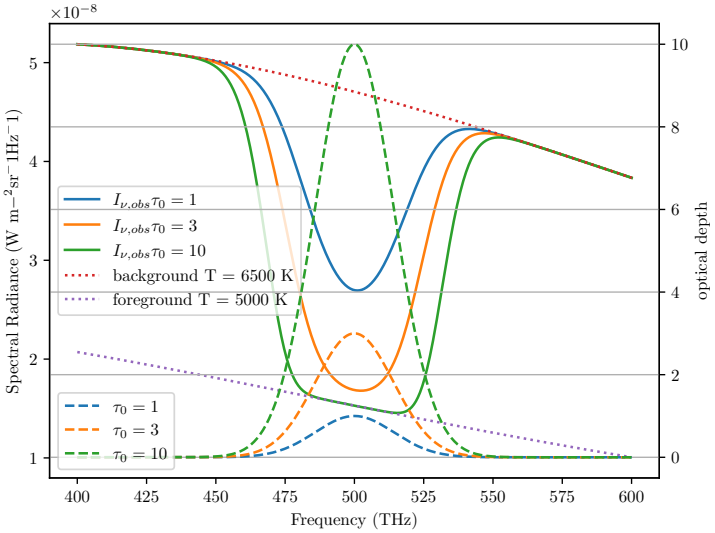


Figure 3.5: An example of absorption features formation for different temperatures and different values of τ. Credits: Prof. Walter del Pozzo.

Local Thermodynamic Equilibrium (LTE)

Let's be honest: In a realistic situation, we *rarely* have strict thermodynamic equilibrium. If a body is in thermodynamic equilibrium, we can assume a number of important physical principles to hold, like the Maxwellian distribution

$$\mathrm{d}n_v = 4\pi n \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right) \mathrm{d}v \tag{3.22}$$

where n is the total number of particles per unit volume and m is the mass of each particle. Similarly, we can expect certain laws to hold, like Boltzmann's law for occupation numbers

$$\frac{n_E}{n_0} = \frac{g_E}{g_0} \exp\left(-\frac{E - E_0}{kT}\right) \tag{3.23}$$

The proverbial "one-million-dollar-question" then is: When can we expect a system to be in thermodynamic equilibrium and when can we expect the previous principles to hold?

Even if the system initially does not obey the, say, Maxwellian distribution, it will eventually relax to it after undergoing some *collisions*.

Collisions are crucial in establishing thermodynamic equilibrium.

When collisions are frequent, the mean free path of particles will be small, and particles will interact more effectively. When this happens, we can expect the principles aforementioned to hold. Since we're physicists, vague sentences like "*the mean free path of particles will be small*" are destined to elicit a deep sense of unease and distress. How small does the free path have to be? One meter? Two micrometers? Below the Planck lengthscale?

When we've defined the absorption coefficient α_ν, the sharpest among my four readers total may have noticed that α_ν has the dimension of the inverse of a length. It is safe to assume that α_ν^{-1} may define some distance over which a significant fraction of the radiation would get absorbed by matter.

Such a "mean-distance" is defined in a homogeneous medium as

$$\langle \tau_\nu \rangle = \alpha_\nu l_\nu = 1$$

Thus, if l_ν is sufficiently small such that the temperature can be taken as a constant over such distance, we can safely say that the useful relations we have defined earlier still hold, although only locally.

In such a fortunate scenario, known as *Local Thermodynamic Equilibrium* (LTE), all the important laws requiring thermodynamic equilibrium are expected to hold, provided that we use the local temperature $T(\vec{x})$.

In the interiors of stars, for example, LTE will prove to be a very good approximation, that will get progressively worse as we approach the "surface" of the star.

Notably, it is possible to show that Kirchhoff's law (3.21) is valid also in the LTE regime.

Then, if we assume LTE, we can use $S_\nu = B_\nu(T)$, so that S_ν can be taken out of the integral (3.17) since the blackbody field is isotropic

$$I_\nu(\hat{k}, \tau_\nu) = I_\nu(\tau_{\nu,0}) \exp(-\tau_\nu) + B_\nu(T)\,(1 - \exp(-\tau_\nu)) \tag{3.24}$$

which is useful to solve in the RJ regime $h\nu/kT \ll 1$ where

$$B_\nu(T) \approx \frac{2k\nu^2}{c^2} T$$

$$I_\nu(\tau_\nu) \overset{def}{=} B_\nu(T_B) \approx \frac{2k\nu^2}{c^2} T_B$$

$$I_\nu(\tau_{\nu,0}) = B_\nu(T_{\mathrm{BG}}) \approx \frac{2k\nu^2}{c^2} T_{\mathrm{BG}}$$

So, in the radio case, in LTE, the radiative transfer solution can be written as

$$\boxed{T_B = T_{\rm BG}\exp(-\tau_\nu) + T\,(1-\exp(-\tau_\nu))} \tag{3.25}$$

where T_B is the observed brightness temperature, $T_{\rm BG}$ is the background brightness temperature, and T is the kinetic temperature of the interstellar material. You should commit this expression to memory, because we're going to use it extensively in the following.

3.6 EINSTEIN COEFFICIENTS AND LINE PROFILE

Kirchhoff's law (eq.3.21), which relates the (spontaneous) emission and the absorption coefficient, seems to imply some underlying microscopic connection between the two phenomena.

As was first discovered by Einstein, that is exactly the case. Let's consider, only for presentation purposes, a simple two-level atom interacting with radiation. As depicted in Fig.3.6, we'll consider two discrete energy levels:

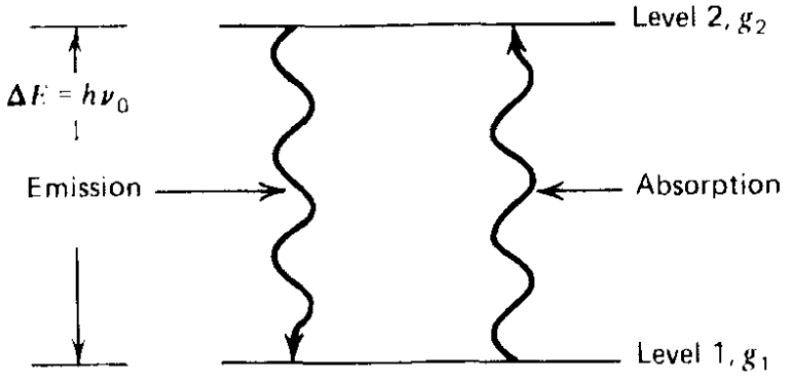


Figure 3.6: Photon emission and absorption in a two levels atom. Credits: [13].

The lower with energy E and degeneracy g_l, while the upper level has energy equal to $E + h\nu_{ul}$ and degeneracy g_u. Transition between the two levels is possible only through absorption ($l \to u$) or emission ($u \to l$) of photons of energy $h\nu_{ul}$.

Three processes can be thus identified: *spontaneous emission*, *stimulated emission* and *absorption*.

Spontaneous Emission

The process we'll refer to as *spontaneous emission* occurs when the system transitions from the excited state u to the lower level l through the emission of a photon.

Spontaneous emission can occur even in the **absence of radiation fields** and can be assumed to be **isotropic**. The transition probability for spontaneous emission is defined through the *Einstein A-coefficient* A_{ul}, which has units $[A_{ul}] = \mathrm{s}^{-1}$.

Absorption

In the presence of a radiation field with the right energy, the system can absorb a photon to transition from state l to a higher energy state. Assuming that the radiation field cannot self-interact, we expect the probability per unit time to be proportional to the density of photons (or to the mean intensity J_ν) at frequency ν_{ul}.

However, the energy difference between the two states is not infinitely sharp, but more like a smoother curve we'll call the *line profile function* $\phi(\nu - \nu_{ul})$

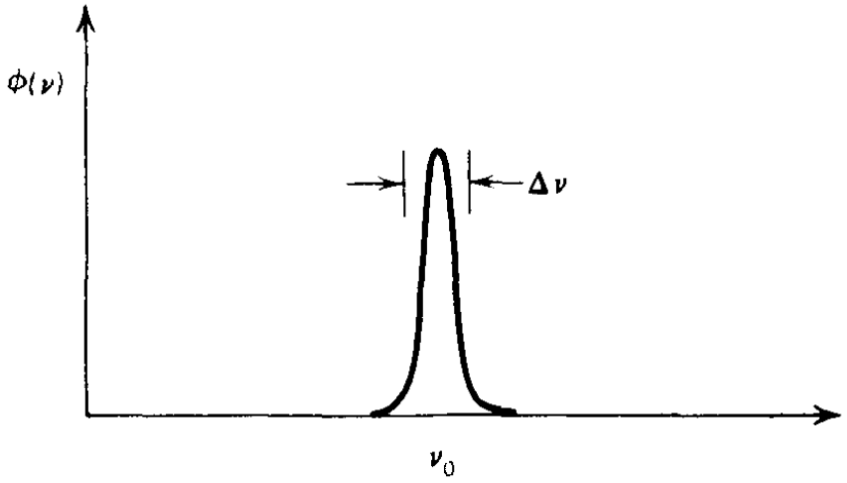


Figure 3.7: Line profile for a two levels atom.
Credits: [13].

which is conveniently peaked at $\nu = \nu_{ul}$ and is correctly normalized

$$\int_0^{+\infty} \phi(\nu - \nu_{ul})\, \mathrm{d}\nu = 1 \tag{3.26}$$

The line profile function is usually a function with the peak on ν_{ul}, and goes rapidly to zero elsewhere. It is useful to also define the *equivalent width*

$$\phi(0)\Delta\nu_{eq} = 1 \tag{3.27}$$

Not bothering for the moment with the underlying physical mechanisms that concur in determining the line profile, we'll propose the following definition for the transition probability for absorption: $B_{lu}\overline{J}$, where $\overline{J}$ is

$$\overline{J} = \int_0^{+\infty} J_\nu \phi(\nu - \nu_{ul})\, \mathrm{d}\nu$$

and B_{lu} is the *Einstein B-coefficient*.

Stimulated Emission

As anticipated, there exists yet another way for a system to emit a photon, but this time requiring the presence of a radiation field. Taken $\overline{J}$ has the

same meaning as before, we'll define the transition probability per unit time for stimulated emission as $B_{ul}\bar{J}$.

If we were to assume that the mean intensity J_ν changes slowly over the width $\Delta\nu$ of the line profile, we could in principle approximate the line profile as a δ-function peaked at ν_0.

Some books use the energy density U_ν in the definitions. It's pretty much the same since they carry the same information, but be aware that the two definitions will differ of a $c/4\pi$ factor.

3.6.1 Line Broadening

It would be most dumb for us to assume that energy levels, or the lines connecting them, are infinitely sharp. Even a quick look at Heisenberg's uncertainty principle should convince you otherwise.

When we defined the Einstein's coefficients, we had to introduce the line profile $\phi(\nu - \nu_{ul})$ to account for the non-zero width of the line, and it's about time we consider some of the phenomena that concur in determining the line's actual shape.

Natural Broadening

For this one, you'll have to recall some of the basics results of Quantum Mechanics. The calculation is not particularly difficult, but it's surely long, so I'm going to omit that and take a simpler route, which is reporting the final result skipping all the calculations.

Note that the spontaneous decay of an atomic state n proceeds at a rate

$$\Gamma = \sum_m A_{nm}$$

where the sum is over all states m of lower energy (which may be quite a few). If radiation is present, we should add the induced rates to this. The coefficient of the wave function of state n, therefore, is of the form $e^{-\Gamma t/2}$ and leads to a decay of the electric field by the same factor. We have an emitted spectrum determined by the decaying sinusoid type of electric field, so the line profile must be a *Lorentz or Natural profile*[4]

$$\phi(\nu - \nu_{ul}) = \frac{\Gamma}{4\pi^2} \frac{1}{(\nu - \nu_{ul})^2 + \left(\frac{\Gamma}{4\pi}\right)^2}$$

More often than not, however, we're going to assume that the effect of natural broadening is rather peaked around ν_{ul}, similar to a δ-function, since, at least as far as astrophysics is concerned, there are more relevant sources of broadening.

4 Fun fact: The Lorentzian distribution (also known as Cauchy distribution among mathematicians) is a classical example of a pathological distribution. In fact the Cauchy distribution has no mean, variance, or higher momenta defined. Its mode and median are well defined and are both equal to ν_0.

Doppler Broadening

Consider an atom in thermal motion, so that by simple relativistic considerations the frequency of emission or absorption in its own frame corresponds to a different frequency for the observer. Each atom will have its own Doppler shift, so that the net effect is to spread the line out, but not to change its total strength.

Recall the classic Doppler effect

$$\nu - \nu_{ul} = \nu_{ul}\frac{v_z}{c} \tag{3.28}$$

Here ν_{ul} is the rest frequency for a given transition between states u and l. Note that this can be extended to *bulk motion* as well. We can decompose the velocity as

$$\vec{v} = \langle \vec{v} \rangle + \delta\vec{v}$$

which goes by the name of *Reynolds' decomposition* in Fluid-dynamics books. Generally, an ensemble of atoms will have different $\delta\vec{v}$, so each atom will absorb photons at different frequencies.

What we have to do is then convolute the Doppler effect over some velocity distribution. Assuming LTE to hold, such distribution will be the Maxwellian distribution. Switching into a reference frame where $\langle \vec{v} \rangle = 0$ (which is always possible), then the small fluctuations $\delta\vec{v}$ shall follow the Maxwellian distribution.

Please note that the one dimensional version of the Maxwellian distribution must be used, which is a Gaussian distribution with $\mu = 0$ and a given variance. The convoluted line profile will then be

$$\phi(\nu - \nu_{ul}) = \int \delta(\nu - \nu_{ul})\nu_{ul}\left(1 + \frac{v}{c}\right)\exp\left(-\frac{m_a v^2}{2kT}\right)\,\mathrm{d}v$$

where, as anticipated, we identified the natural broadening as a δ-function. This is easily computed

$$\phi(\nu - \nu_{ul}) = \frac{1}{\Delta\nu_D\sqrt{\pi}}e^{-(\nu-\nu_{ul})^2/(\Delta\nu_D)^2} \quad \Delta\nu_D = \frac{\nu_{ul}}{c}\sqrt{\frac{2kT}{m_a}} \tag{3.29}$$

Here $\Delta\nu_D$ represents the *Doppler (thermal) width*.

Generally speaking, the minimum broadening we can expect is given by a convolution of Doppler Broadening and Natural Broadening in its proper Lorentzian form. This hasn't a fancy analytical solution, but can be expressed in terms of what is called a *Voigt profile*.

Roughly speaking, the Voigt profile is something that looks like a Gaussian at the center of the line[5] (where Doppler effect dominates) but has the *wings* of the Lorentzian distribution.

$$H(a,u) = \frac{a}{\pi}\int_{-\infty}^{+\infty}\frac{e^{-y^2}}{a^2 + (u-y)^2}\,\mathrm{d}y \quad a = \frac{\Gamma}{4\pi\Delta\nu_D} \quad u = \frac{\nu - \nu_0}{\Delta\nu_D}$$

In terms of this compact definition of the Voigt profile, the overall line profile will be

$$\phi(\nu - \nu_{ul}) = \frac{1}{\Delta\nu_D\sqrt{\pi}}H(a,u)$$

5 Sometimes it's called *kernel* of the distibution.

Note that in principle we also have a non-thermal Doppler contribution coming from turbulent motion and systematic motion of the fluid we're considering. If all terms are Gaussian-distributed (which is an assumption that might need some care), the total width of the line due to Doppler effects is simply

$$\Delta\nu^2_{\text{obs}} = \Delta\nu^2_{\text{th}} + \Delta\nu^2_{\text{turb}} + \Delta\nu^2_{\text{syst}}$$

3.7 RELATIONS BETWEEN THE EINSTEIN COEFFICIENTS

It would be most useful to know if there were some kind of relations between the three different coefficients, and if there were some way of relating them to what we've called α_ν and j_ν when dealing the radiative transport.

If this weren't the case, this section would have no reason to exist, so it's safe for you to assume that such relations do in fact exist. To do so, however, we have to invoke the *quantum theorem of detailed balance* (see, for example, [7]), which roughly says

> If states a and b of a system have the same energy, then if P_{ab} is the probability per unit time of a transition from a to b, and P_{ba} from b to a,
>
> $$P_{ab} = P_{ba}$$

For an atom of matter to be in equilibrium with a radiation field, then, the probability of said atom being in the ground state and absorbing a photon of a given frequency must be equal to the probability that it is in the excited state and emits the photon.

If we assume to be in a steady-state condition, the rate of transition from level 1 to level 2 has to be equal to the rate of transition from level 2 to level 1, or, more generally

$$n_i \sum_j R_{ij} - \sum_j n_j R_{ji} = 0$$

If we put in the transition rates we've written down earlier, we get for the two-level atom

$$n_l B_{lu} \bar{J} = n_u A_{ul} + n_u B_{ul} \bar{J}$$

which means that the number of transitions per unit time per unit volume out of state l must be equal to the number of transitions per unit time per unit volume into state l.

Solving for $\bar{J}$

$$\bar{J} = \frac{n_u A_{ul}}{n_l B_{lu} - n_u B_{ul}} = \frac{A_{ul}/B_{ul}}{(n_l/n_u)\cdot(B_{lu}/B_{ul}) - 1} \tag{3.30}$$

In thermodynamic equilibrium we can use eq.3.23 to relate the occupation numbers

$$\frac{n_l}{n_u} = \frac{g_l}{g_u} \frac{\exp(-E/kT)}{\exp[-(E+h\nu_{ul})/kT]} = \frac{g_l}{g_u} \exp(h\nu_{ul}/kT)$$

but in thermodynamic equilibrium we know that $J_\nu = B_\nu$, so if B_ν varies slowly on the scale of $\Delta\nu$ we can assume $\bar{J} \approx B_\nu(\nu_0)$.

That means the following relations must simultaneously hold

$$g_l B_{lu} = g_u B_{ul} \tag{3.31}$$

$$A_{ul} = \frac{2h\nu_{ul}^3}{c^2} B_{ul} \tag{3.32}$$

which are known as *Einstein relations*. They connect atomic properties A_{ul}, B_{ul} and B_{lu} and have no reference to the temperature T (unlike Kirchhoff's law).

Although we have derived eq.3.32 assuming thermodynamic equilibrium, those two relations **must be always valid** whether or not the atoms are in thermodynamic equilibrium.

3.7.1 Absorption and Emission coefficients in terms of Einstein coefficients

To obtain the emission coefficient j_ν we have to make a crucial assumption about the frequency distribution of the emitted radiation: This emission is distributed with the same line profile $\phi(\nu - \nu_{ul})$ that describes absorption, which is often verified in astrophysics (good for us).

By the definition (3.11), we already know the amount of energy emitted in volume $\mathrm{d}V$, solid angle $\mathrm{d}\Omega$, frequency $\mathrm{d}\nu$, and time $\mathrm{d}t$. Now, since each atom contributes with $h\nu_{ul}$ to spontaneous emission over a 4π solid angle, we can express the emission coefficient as

$$j_\nu = \frac{h\nu_{ul}}{4\pi} n_u A_{ul} \phi(\nu - \nu_{ul}) \tag{3.33}$$

Similarly, you can show that the energy absorbed from radiation in frequency range $\mathrm{d}\nu$, solid angle $\mathrm{d}\Omega$, time $\mathrm{d}t$ and volume $\mathrm{d}V$ is

$$\frac{h\nu_{ul}}{4\pi} n_l B_{lu} I_\nu \phi(\nu - \nu_{ul}) \, \mathrm{d}\nu \, \mathrm{d}\Omega \, \mathrm{d}t \, \mathrm{d}V$$

From here follows immediately the *true absorption coefficient*

$$\alpha_\nu = \frac{h\nu_{ul}}{4\pi} n_l B_{lu} \phi(\nu - \nu_{ul}) \tag{3.34}$$

Note that since stimulated emission is proportional to the specific intensity I_ν and only affects the photons along the given beam, it's not a big stretch to claim that it behaves in a way that is akin to true absorption. We shall then define the *absorption coefficient, corrected for stimulated emission* as

$$\alpha_\nu = \frac{h\nu_{ul}}{4\pi} \phi(\nu - \nu_{ul})(n_l B_{lu} - n_u B_{ul}) \tag{3.35}$$

in which we're regarding stimulated emission like a *negative absorption*.

In terms of the newly defined emission and absorption coefficients, the equation of radiative transfer takes a new look

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = -\frac{h\nu_{ul}}{4\pi} \phi(\nu - \nu_{ul})(n_l B_{lu} - n_u B_{ul}) I_\nu + \frac{h\nu_{ul}}{4\pi} n_u A_{ul} \phi(\nu - \nu_{ul}) \tag{3.36}$$

Computing the ratio j_ν / α_ν with the new definitions holds

$$S_\nu = \frac{n_u A_{ul}}{n_l B_{lu} - n_u B_{ul}} = \frac{2h\nu_{ul}^3}{c^2} \left(\frac{g_u n_l}{g_l n_u} - 1 \right)^{-1} \tag{3.37}$$

where the last equality is obtained just by substitution of the Einstein relations (3.32). Note that eq.(3.37) is a generalized Kirchhoff's law which holds also for nLTE configurations.

Recall the definition (3.35), which under the assumption of LTE turns into

$$\alpha_\nu = \frac{h\nu_{ul}}{4\pi} n_u B_{ul} \left[\exp\left(\frac{h\nu_{ul}}{kT}\right) - 1 \right] \phi(\nu - \nu_{ul})$$

where we've made use of (3.23) to relate n_u and n_l and of Einstein's relations (3.32). Integrating over the line of sight, we can also calculate the optical depth from the definition (3.15)

$$\tau_{nu} = -\int \frac{h\nu_{ul}}{4\pi} n_u B_{ul} \left[\exp\left(\frac{h\nu_{ul}}{kT}\right) - 1 \right] \phi(\nu - \nu_{ul})\, \mathrm{d}s \tag{3.38}$$

which, if we assume temperature to be roughly constant along the line of sight, simplifies to

$$\tau_\nu \propto \int n_u \, \mathrm{d}s = N_u \tag{3.39}$$

where we've defined the *column density* of the level u.

3.8 STATISTICAL EQUILIBRIUM

So far, the assumption of LTE has trailed along our whole discussion like a ghost, which prevented us from looking back by spooking us with creepy-sounding sentences like: "What if LTE doesn't hold?"

However, as our favorite cartoon from our kid-days taught us, below the mask of the ghost there usually is the solution to all our problems. Let's see then how we can tackle the matter and chase the ghost away.

In nLTE Boltzmann's law (3.23) clearly doesn't hold. But what if we're in such a scenario that *statistical equilibrium*[6] holds instead? Well, in that case we can define a temperature called *excitation temperature* T_{ex} that is such that

$$\frac{n_u}{n_l} = \frac{g_u}{g_k} \exp\left(-\frac{h\nu_{ul}}{kT_{\mathrm{ex}}}\right) \tag{3.40}$$

where we're implicitly assuming the ratio n_u/n_l to be constant over time. Note that if the system is in LTE, then $T_{\mathrm{ex}} = T$ and is the same $\forall(u,l)$, while in nLTE T_{ex} is different for each transition considered.

In statistical equilibrium, the source function S_{nu} can then be written as

$$S_\nu = \frac{j_\nu(T_{\mathrm{ex}})}{\alpha_\nu(T_{\mathrm{ex}})} = B_\nu(T_{\mathrm{ex}}) \tag{3.41}$$

and the optical depth can be defined as well

$$\tau_{nu} = -\int \frac{h\nu_{ul}}{4\pi} n_u B_{ul} \left[\exp\left(\frac{h\nu_{ul}}{kT_{\mathrm{ex}}}\right) - 1 \right] \phi(\nu - \nu_{ul})\, \mathrm{d}s \tag{3.42}$$

6 Statistical equilibrium: "A state in a system where the overall probability distribution of its states or macroscopic properties remains constant over time, even as individual micro-level changes or fluctuations continue to occur." Credits: this website.

Consider now a system in statistical equilibrium comprising of a background source of radiation $I_\nu^{\rm BG}$ assumed to emit like a blackbody[7]. The brightness of a line will be the difference between the total brightness at a given frequency and the background brightness

$$I_\nu^{\rm line} = I_\nu^{\rm tot} - I_\nu^{\rm BG} = I_\nu^{\rm tot} - B_\nu^{\rm BG} \tag{3.43}$$

Making use of the expansion of the radiative transfer equation in the RJ regime for $I_\nu^{\rm tot}$, we can write, exploting the definition we've given of excitation temperature

$$\begin{aligned} I_\nu^{\rm line} &= B_\nu(T_{\rm BG})e^{-\tau_\nu} + B_\nu(T_{\rm ex})(1-e^{-\tau_\nu}) - B_\nu(T_{\rm BG}) \\ &= \left[B_\nu(T_{\rm ex}) - B_\nu(T_{\rm BG})\right]\left(1-e^{-\tau_\nu}\right) \end{aligned}$$

Now, with the further definition of line brightness temperature T_B so that $I_\nu^{\rm line} = B_\nu(T_B)$, and under the assumption of being in the RJ regime

$$\boxed{T_B = (T_{\rm ex} - T_{\rm BG})\left(1 - \exp(-\tau_\nu)\right)} \tag{3.44}$$

From this equation is particularly simple to show that "narrow" lines are better seen in absorption rather than in emission. In particular, we're going to say that

- If $T_{\rm ex} > T_{\rm BG}$, the line is an *emission line*;
- If $0 \leq T_{\rm ex} < T_{\rm BG}$, the line is an *absorption line*;
- If $T_{\rm ex} < 0$, the line is a *maser line*;

3.8.1 Maser lines

Consider lines that originate in regions where particle have undergone the *(population) inversion process* $n_u > n_l$. This means

$$\frac{n_u}{n_l} = \frac{g_u}{g_l}\exp\left(-\frac{h\nu_{ul}}{kT_{\rm ex}}\right) \quad n_u > n_l \iff T_{\rm ex} < 0 \tag{3.45}$$

This phenomenon is mostly seen in species with two excited levels A, and B with slight energetic differences and a third level C with much lower energy. Whenever the transition C→A is favored with respect to C →B, a population inversion might occur.

from which

$$\tau_{nu} = -\int \frac{h\nu_{ul}}{4\pi} n_u B_{ul}\left[\exp\left(\frac{h\nu_{ul}}{kT_{\rm ex}}\right) - 1\right]\phi(\nu - \nu_{ul})\,{\rm d}s < 0 \tag{3.46}$$

if $T_{\rm ex} < 0$. If we plug this same condition in (3.44) we get

$$T_B = (|T_{\rm ex}| + T_{\rm BG})\left[\exp\left(|\tau_\nu|\right) - 1\right]$$

which describes an *amplification* and not an attenuation.

A MASER (*Microwave Amplification by Stimulated Emission of Radiation*) is then easily identified by mainly three characteristics

- Huge line temperature T_B;

7 It might be, for example, the microwave emission of the CMB, which is the best example of blackbody emission.

- Point-like source: Maser emission has a small angular size, typically in coherent "spots" < 1 mas;

- High temporal variability (Fig.3.8): The exponential dependence on τ reflects into a strong dependence even on small variations of the relative population of the levels and on the amplification path.

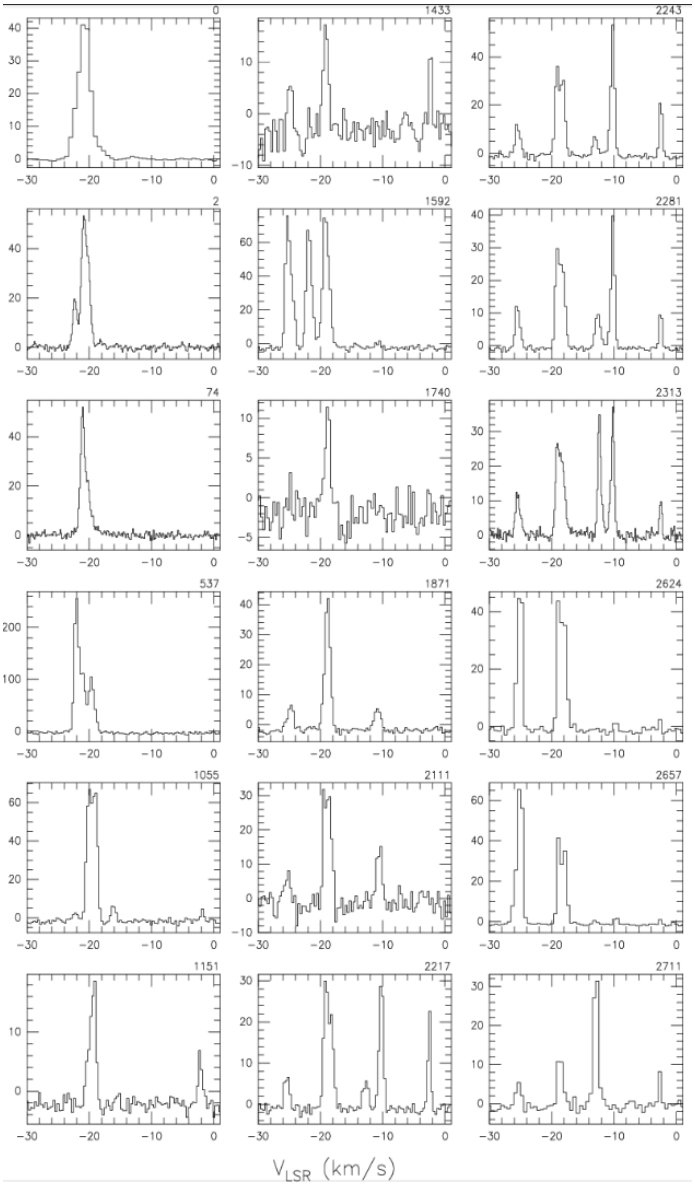


Figure 3.8: Variability of the H_2O maser in the star forming region L1204-G.

Maser lines are particularly useful for two reasons: (a) They're typical of star forming regions, (b) they're really bright, albeit extremely spatially confined.

Maser lines are mainly divided into two classes: *Collisionally excited lines*, and *radiatively excited lines*.

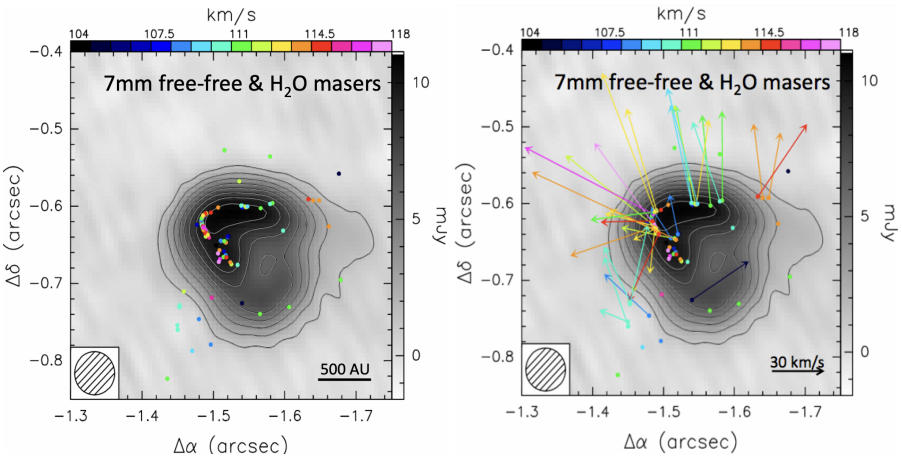


Figure 3.9: Analysis of 7 mm free-free radiation lines and H_2O maser lines in a H_{II} region. The panel on the right shows that the hypercompact H_{II} region seems to be (generally) expanding. For more you can check [11].

3.9 EXCITATION TEMPERATURE AND COLUMN DENSITY

For this, we'll have to make extensive use of (3.44). For the sake of simplicity, let's assume $T_{ex} \gg T_{BG}$. There are two cases worth exploring: (a) the case of the optically thin medium, (b) the case of the optically thick medium.

For the latter, a simple Taylor-expansion tells us that

$$T_B \approx T_{ex}\tau_\nu$$

but since

$$\tau_{nu} = \frac{h\nu_{ul}}{4\pi}(N_l B_{lu} - N_u B_{ul})\phi(\nu - \nu_{ul})$$

in statistical equilibrium. We can then integrate the brightness temperature over frequencies, thus finding

$$\int_0^\infty T_B \, d\nu \approx \int_0^\infty T_{ex}\tau_\nu \, d\nu = T_{ex}\frac{h\nu_{ul}}{4\pi}(N_l B_{lu} - N_u B_{ul})$$

where we've exploited the normalization property of the line profile (as well as assuming that the excitation temperature is independent on the frequency). From the relation between the Einstein coefficients and the definition of excitation temperature (3.40), we have

$$\int_0^\infty T_B \, d\nu = \frac{T_{ex}}{4\pi} N_u \frac{c^2}{2\nu_{ul}^2} A_{ul}\left[\exp\left(\frac{h\nu_{ul}}{kT_{ex}}\right) - 1\right] \tag{3.47}$$

In Rayleigh-Jeans approximation, the former equation becomes

$$\int_0^\infty T_B \, d\nu = \frac{h}{4\pi k} N_u \frac{c^2}{2\nu_{ul}} A_{ul}$$

from which, transforming the integral in frequency to integral to velocities we get an equation for the column density N_u

$$\boxed{N_u \approx \frac{8\pi k \nu_{ul}^2}{hc^3 A_{ul}} \int_{-\infty}^{+\infty} T_B \, dv} \tag{3.48}$$

Note that we're already assuming T_{ex} to be omogeneous along the line of sight, so that the exponential term in (3.42) can come out of the integral, which simply turns into the definition of column density.

The case for the optically thick medium is perhaps even simpler. In fact

$$T_B = (T_{\rm ex} - T_{\rm BG})(1 - \exp(-\tau_\nu)) \sim T_{\rm ex}$$

In LTE, $T_{\rm ex} = T_k \implies T_B = T_k$. So, the brightness temperature of an optically thick line gives a direct estimate of $T_{\rm ex}$, but not of the column density.

3.9.1 Boltzmann Diagrams

In LTE we know that

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} \exp\left(-\frac{E_u - E_l}{kT}\right)$$

By summing both members of Boltzmann equation on all l, we obtain

$$\frac{n_u}{g_u} = \frac{n_{\rm tot}}{\mathcal{Z}(T)} \exp\left(-\frac{E_u}{kT}\right)$$

where

$$n_{\rm tot} = \sum_l n_l \quad \mathcal{Z} = \sum_l g_l \exp\left(-\frac{E_l}{kT}\right)$$

By integrating n_u and $n_{\rm tot}$ over the line of sight, we can define the column density for the state u and the total column density, so that plugging them in the Boltzmann equation, and expressing it through logarithm

$$\ln\left(\frac{N_u}{g_u}\right) = \ln\left(\frac{N_{\rm tot}}{\mathcal{Z}(T)}\right) - \frac{E_u}{kT} \tag{3.49}$$

On a $x = E_u$ and $y = \ln(N_u/g_u)$ plot, this is the equation of a straight line!

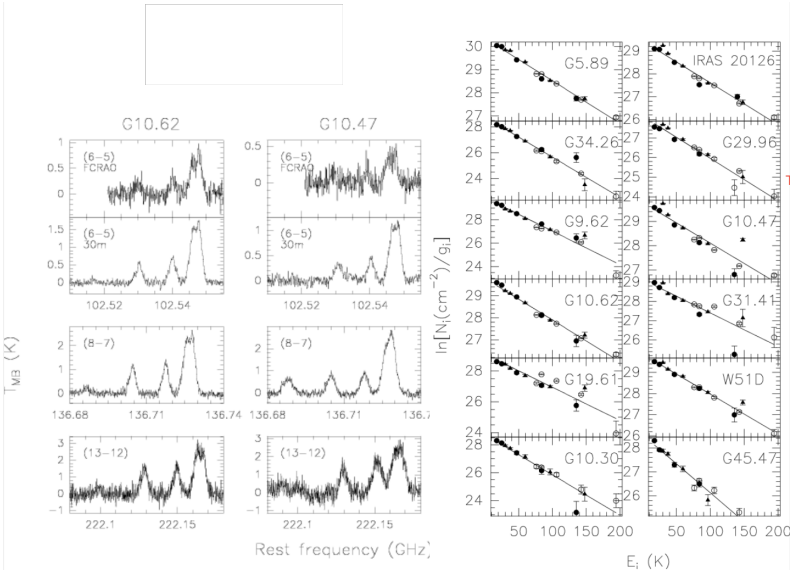


Figure 3.10: Rotational transitions of methyl-acetylene CH_3C_2H observed in molecular clumps associated with 12 ultracompact (UC) H_{II} regions. On the right, the associated Boltzmann diagrams. Credits: [6].

3.10 NLTE AND RATE OF COLLISIONAL TRANSITIONS

As you might have guessed, collisions are fairly important in astrophysics. For our purposes it is worth recalling that particle transitions in the ISM depend both on the interaction with a radiation field, and on collisions with other particles.

First, let us find a way to parametrise the rate of collisional transitions similar to radiative transitions. Remember that the collisional rate in a gas with Maxwellian velocity distribution, i.e. the collisions per unit time in the unit volume, is given by

$$\Gamma_{up} = n_u n_p \langle \sigma_{ul} v \rangle$$

where n_p is the volume density of the collisional partners; in pratice, it will always be some species of Hydrogen. We define as *collisional coefficient* the quantity

$$C_{ul} = n_p \langle \sigma_{ul} v \rangle \tag{3.50}$$

so that $\Gamma_{ul} = n_u C_{ul}$.

Let's now consider a stationary state, so non necessarily in LTE. The net number of transitions to the u level from any other l level per unit volume and time must be vanishing by principle of the detailed balance principle

$$n_u \left[\sum_{l \neq u} (C_{ul} + B_{ul} J_\nu) + \sum_{u>l} A_{ul} \right] = \sum_{l \neq u} (n_l B_{lu} J_\nu + n_l C_{lu}) + \sum_{u<l} n_l A_{ul} \tag{3.51}$$

which, of course, is more easily discussed for the two-level system.

Let's assume then that only two levels, 1 and 2, are populated. In practice, any system in which the exchange between the two levels considered and a third one can be neglected, can be approximated as a two level system. We already know that

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} \exp\left(-\frac{h\nu_{21}}{kT_{\rm ex}}\right) \implies \frac{C_{21}}{C_{12}} = \frac{g_2}{g_1} \exp\left(-\frac{h\nu_{21}}{kT_{\rm ex}}\right) \tag{3.52}$$

the implication stems from the principle of detailed balance, which is valid if collisional partners are in LTE.

The statistical equilibrium equations (3.51) then simplify to just

$$n_2(A_{21} + B_{21} J_\nu + C_{21}) = n_1(B_{12} J_\nu + C_{12}) \tag{3.53}$$

If there's no radiation field $J_\nu = 0$, the equation is easily solved

$$\begin{aligned} & n_1 C_{12} = n_2 C_{21} + n_2 A_{21} \\ \implies & \frac{n_2}{n_1} = \frac{C_{12}}{A_{21}} \left[1 + \frac{C_{21}}{A_{21}}\right]^{-1} = \frac{C_{12}}{C_{21}} \frac{C_{21}}{A_{21} + C_{21}} \end{aligned}$$

which is also equal to

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} \exp\left(-\frac{h\nu_{21}}{kT}\right) \frac{1}{1 + A_{21}/C_{21}} \tag{3.54}$$

We define the *critical density* n_c as

$$n_c = \frac{A_{21} n_p}{C_{21}} \tag{3.55}$$

which is a *specific of the transition that we're considering*. In terms of the critical density

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} \exp\left(-\frac{h\nu_{21}}{kT}\right) \frac{1}{1 + n_c/n_p} \tag{3.56}$$

This last equation admits two limits

$$n_p \gg n_c \implies \frac{n_2}{n_1} \approx \frac{g_2}{g_1} \exp\left(-\frac{h\nu_{21}}{kT}\right) \tag{3.57}$$

$$n_p \gg n_c \implies \frac{n_2}{n_1} \approx \frac{n_p}{n_c}\frac{g_2}{g_1} \exp\left(-\frac{h\nu_{21}}{kT}\right) \tag{3.58}$$

so in the latter case a further suppression is introduced to the Boltzmann law for occupation numbers.

What if $J_\nu \neq 0$? The calculation is a lot messier, and algebraically-intense, but not particularly different. We're still exploting Einstein's equations, the definition of excitation temperature, and the fact that the background radiation is (likely) a blackbody.

The final result is

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} \frac{\left[\exp(h\nu/kT_{\rm BG}) - 1\right]^{-1}}{1 + \left[\exp(h\nu/kT_{\rm BG}) - 1\right]^{-1} + n_p/n_c} + \\ + \frac{g_2}{g_1} \frac{\exp\left(-h\nu/kT\right)}{1 + \frac{n_c}{n_p}\left(1 + \left[\exp(h\nu/kT_{\rm BG}) - 1\right]^{-1}\right)}$$

where perhaps is useful to introduce the *generalized critical density* for the transition

$$n_c^* = n_c \left[1 + \frac{1}{\exp(h\nu/kT_{\rm BG}) - 1}\right] \tag{3.59}$$

through which we write

$$\boxed{\frac{n_2}{n_1} = \frac{g_2}{g_1} \frac{1}{\exp(h\nu/kT_{\rm BG}) - 1} \frac{n_c}{n_c^* + n_p} + \frac{g_2}{g_1} \frac{\exp\left(-h\nu/kT\right)}{\left(1 + n_c^*/n_p\right)}} \tag{3.60}$$

In the limit $n_c^* \gg n_p$

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} \exp\left(-h\nu/kT_{\rm ex}\right) \approx \frac{g_2}{g_1} \exp\left(-h\nu/kT_{\rm BG}\right) \implies T_{\rm ex} \sim T_{\rm BG} \tag{3.61}$$

This limit defines a *sub-thermal line*, which is the one satisfying

$$T_B = (T_{\rm ex} - T_{\rm BG})(1 - \exp(-\tau_\nu)) \approx 0 \tag{3.62}$$

while the opposite limit $n_c^* \ll n_p$ defines a *thermal line*

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} \exp\left(-h\nu/kT_{\rm ex}\right) \approx \frac{g_2}{g_1} \exp\left(-h\nu/kT\right) \implies T_{\rm ex} \sim T \tag{3.63}$$

which verifies

$$T_B = (T_{\rm ex} - T_{\rm BG})(1 - \exp(-\tau_\nu)) \approx (T - T_{\rm BG})(1 - \exp(-\tau_\nu)) \tag{3.64}$$

3.11 OBSERVATIONS OF MOLECULAR CLOUDS AND SF REGIONS

Chances are that for our observations we're going to use a telescope of some kind, be it a single-dish one or an interferometric one. Neither of the two, however, are able to gauge directly the brightness temperature of the observed source, but rather measure the convolution between the response of the instrument itself and the temperature, proper, of the source.

It is common then to introduce the quantity $P_n(\theta, \phi)$ known as *normalized antenna diagram*, which is the equivalent of the PSF in the optical bands. In polar coordinates, it represents the power received as a function of the angular distance $\theta - \theta_0$, $\phi - \phi_0$ from the pointing direction.

Since telescopes operate through diffraction processes, it is expected of them to produce multiple "lobes," which essentially tell us where the telescope is most sensitive. The larger lobe is usually called *main lobe* or *main beam*, and the solid angle it subtends is called Ω_{MB}, the main beam solid angle.

Let's assume now that an astrophysical source is emitting brightness $I_\nu = B_\nu(T_B)$, parametrized with the brightness temperature. In the RJ-regime, we can write

$$B_\nu(T) \approx \frac{2k\nu^2}{c^2} T \implies I_\nu = \frac{2k\nu^2}{c^2} T_B$$

and the telescope's output is the convolution between T_B and P_n, defined as *main beam brightness temperature* T_{MB}

Note that if T_B is a δ-function and P_n is Gaussian, then T_{MB} is Gaussianly distributed as well.

$$T_{MB}(\theta_0, \phi_0) = \frac{\iint T_B(\theta, \phi) P_n(\theta - \theta_0, \phi - \phi_0)\, \mathrm{d}\Omega}{\iint P_n(\theta, \phi)\, \mathrm{d}\Omega} \tag{3.65}$$

One basilar diffraction result is that the angular resolution of a single-dish telescope in the radio/millimetric bands is diffraction-limited, hence the half-power bandwidth is

$$\mathrm{HPBW} = \frac{1.2\lambda}{D} \tag{3.66}$$

which is also the reason why single-dish radio telescopes have (generally) an inferior resolution compared with an equivalent size telescope in the optical bands.

Assuming a Gaussian, point-like source $\Omega_S \ll \Omega_{MB}$

Note that for an "extended source" $T_{MB} \approx T_B$, while for a point-like source $T_{MB} \ll T_B$.

$$T_{MB} = \frac{T_B \Omega_S}{\Omega_{MB} + \Omega_S} \approx \frac{T_B \Omega_S}{\Omega_{MB}} \tag{3.67}$$

If we now substitute the expression we've derived for the brightness temperature (3.44), we obtain

$$T_{MB} = \frac{\Omega_S}{\Omega_{MB} + \Omega_S} (T_{\mathrm{ex}} - T_{\mathrm{BG}})(1 - \exp(-\tau_\nu)) \tag{3.68}$$

which holds under the aforementioned assumptions. Starting from three simple observables (T_{MB}, ν, and Ω_S), quite a plethora of derived quantities can be inferred:

- The excitation temperature T_{ex},

- The optical depth τ;
- The velocity field v;
- The kinetic temperature T;
- The total column density of a given species N_X;
- The mass of molecular Hydrogen M_{H_2};
- The numeric density of molecular Hydrogen n_{H_2}.

Velocity Field

The velocity field is perhaps one of the (conceptually) easier ones to measure. All you have to do is simply infer the velocity of the source from observations of the line profile exploiting the Doppler effect

$$v = \frac{c}{\nu_0}(\nu_0 - \nu) \tag{3.69}$$

Note that in most cases the purely thermal broadening is negligible compared to the total broadening that is observed, which means that the primary sources of broadening are rather turbulence and ordered motions; also, in principle any molecule can be used.

Different ordered motions are dominant during different phases of the evolution of a star-forming region [14]. For example, during the *fragmentation* and *pre-stellar* phases gas contraction and gas infall are the dominant motions, while in the *proto-stellar* phase the dominating contributions are rather those coming from the accretion disk, from outflows, jets, and infalls of gas. Lastly, in the *pre-main sequence* phase, there's only a residual contribution from jets, with most of the contribution coming from disks.

Assume for example that you're trying to observe the emission line of a certain molecule in a rotating disk, as pictorially shown in Fig.3.11.

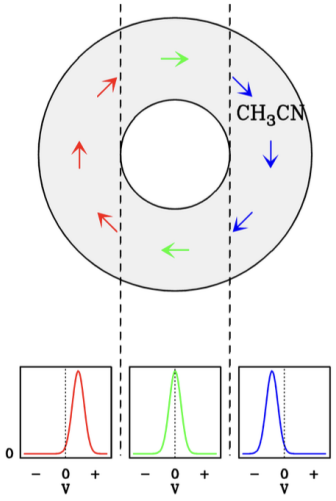


Figure 3.11: Doppler shift on an emission line of some molecule caused by emission in a rotating disk or ring. The observer is supposed to be located somewhere near the bottom of the figure.

A well-established method to resolve small velocity structures is that of the *channel maps*, which is able to give you a feeling of which part of the cloud is moving towards or away from you. One notable application of this method was given by Guilloteau [8] to study emission lines in the young quadruple system GG Tau.

Bipolar Outflows and Infalls

Another rather common occurrence is for a star-forming core to present two outflowing jets, whence the name *bipolar outflow*. Clearly, since they're going to be ejected in diametrally-opposite directions, one of the jets will be blueshifted, while the other will be redshifted, thus modifying the emission profile by adding high-velocity line wings, as shown in Fig.3.12.

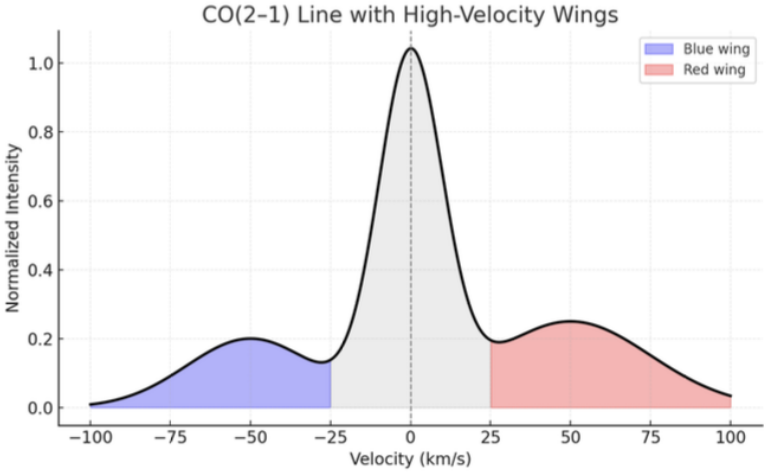


Figure 3.12: Formation of high-velocity line wings induced by the presence of bipolar outflows in a star-forming region.

Note that if the outflow is not spatially resolved, it can be revealed by an excess emission in the high-velocity line wings. Also note that high-velocity wings tend to grow more and more clear for lines with large E_u.

If, on the other hand, there's an instance of infalling envelope, we might expect to see something like what's shown in Fig.3.14.

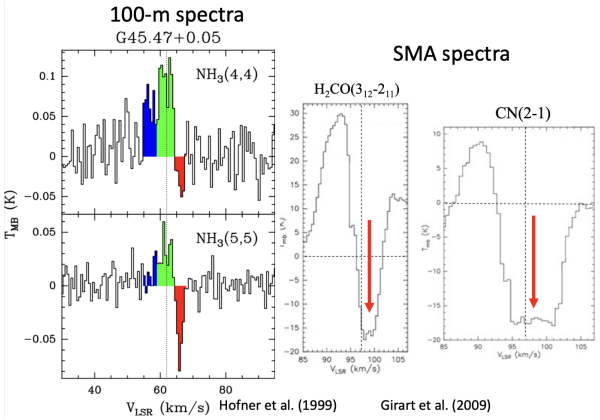


Figure 3.13: Examples of Inverse P-Cygni profiles for different molecules emission lines.

The net result is that different regions of the envelope will be blueshifted, redshifted, or not shifted at all, creating a rather peculiar spectrum.

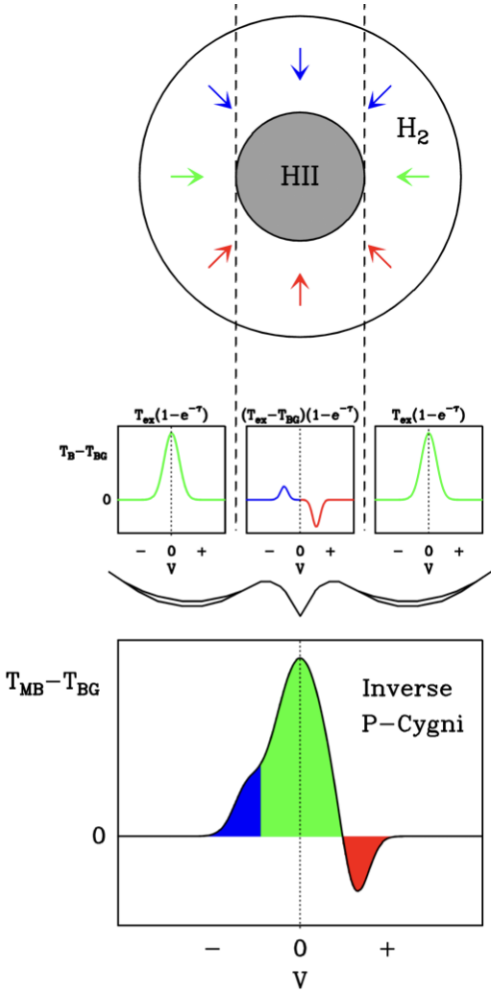


Figure 3.14: Line profile as modified by an infalling envelope. In particular, the one in the picture is an envelope of molecular Hydrogen being accreted onto a H_{II} region.

3.12 OPTICAL DEPTH MEASUREMENTS

It's easy to show that from the same transition observed in two different isotopologues, having known their relative abundance and under the assumption of $T_{BG} \ll T_{ex}$

$$T^a_{MB} = \frac{\Omega_S}{\Omega_S + \Omega_{MB}} T^a_{ex}(1 - \exp(-\tau_a))$$
$$T^b_{MB} = \frac{\Omega_S}{\Omega_S + \Omega_{MB}} T^b_{ex}(1 - \exp(-\tau_b))$$

which tells us that

$$\frac{T^a_{MB}}{T^b_{MB}} \approx \frac{1 - \exp(-\tau_a)}{1 - \exp(-\tau_b)} \tag{3.70}$$

Moreover, since

$$\frac{\tau_b}{\tau_a} = \frac{N_b}{N_a} = X_b \implies \frac{T^a_{MB}}{T^b_{MB}} \approx \frac{1-\exp(-\tau_a)}{1-\exp(-\tau_a X_b)} \tag{3.71}$$

From a line with hyperfine structure, comparing relative intensity of two spectrally resolved components allows to compute of the main one. Same as above, but X_b is replaced by g, the ratio of the two levels, g_2/g_1.

Note that if the system is in LTE (and so $T_{\rm ex} = T$), we can infer two important quantities from just two of our observables (Ω_S and T_{MB}), which are

$$(\tau \gg 1) \quad T \approx \frac{\Omega_{MB}+\Omega_S}{\Omega_S} T_{MB} \tag{3.72}$$

and

$$(\tau \ll 1) \quad N_u \propto \frac{\Omega_{MB}+\Omega_S}{\Omega_S} T_{MB} \tag{3.73}$$

which allows us to create Boltzmann diagrams to infer also the total numeric density (assuming the specifics of the analysed transition are known).

If, instead, we're in a nLTE system, we can still calculate the numeric density of a given excited state from a single line with $\tau \ll 1$ (and under the assumption of the same excitation temperature for all rotational levels)

$$N_u = \frac{8\pi k \nu^3}{hc^3 A_{ul}} \frac{\Omega_{MB}+\Omega_S}{\Omega_S} \int_{-\infty}^{\infty} T_{MB}\, \mathrm{d}\nu \tag{3.74}$$

and infer the total numeric density

$$N_{\rm tot} = \frac{N_u \mathcal{Z}(T_{\rm ex})}{g_u} \exp(E_u/kT_{\rm ex}) \tag{3.75}$$

With more than one line we could instead use Boltzmann diagrams.

If the system is optically thick ($\tau \gg 1$) to the transition, from the analysis of one line only we get

$$N_u = \frac{8\pi k \nu^3}{hc^3 A_{ul}} \frac{\Omega_{MB}+\Omega_S}{\Omega_S} \int_{-\infty}^{\infty} T_{MB} \frac{\tau_\nu}{1-\exp(-\tau_\nu)}\, \mathrm{d}\nu \tag{3.76}$$

while with at least two lines we can still create Boltzmann diagrams.

It is not particularly difficult to show that the numeric density N_u and the numeric density under the assumption of optical thinness are related by a simple expression

$$N_u^{\rm thin} = \frac{1-\exp(-\tau^0)}{\tau^0} N_u \tag{3.77}$$

with τ^0 defined through the introduction of the equivalent width $\Delta\nu$

$$\int \tau_\nu \, \mathrm{d}\nu = \tau^0 \Delta\nu \tag{3.78}$$

3.13 MEASUREMENTS OF DENSITIES AND MASSES

In the following, we'll try to give a short account of how measurements of masses and densities of species of astronomical interest are possible thanks to what we've discussed so far.

3.13.1 Density of Molecular Hydrogen

From the total numeric density $N_{\rm tot}$ we can calculate the numeric density of molecular Hydrogen

$$n_{\rm H_2} = \frac{N_{\rm tot}}{D[X]} \tag{3.79}$$

where D is the source's diameter, and X is the molecular abundance with respect to molecular Hydrogen.

3.13.2 Dust and Gas Mass

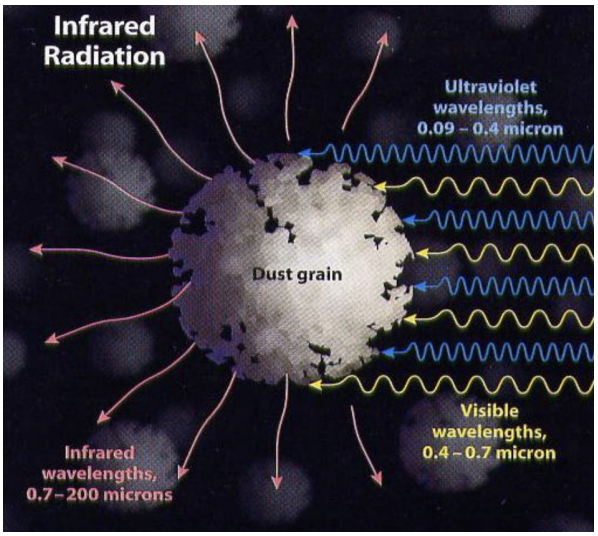


Figure 3.15: Schematic representation of emission and absorption as carried out by dust grains in the ISM.

The flux density of a source with thermal emission from dust, at temperature T_d and solid angle Ω_S, is

$$F_\nu = \int_{\Omega_S} I_\nu \, {\rm d}\Omega = B_\nu(T_d)(1 - \exp(-\tau_\nu))\Omega_S \tag{3.80}$$

where the factor multiplying Ω_S is called *grey body*[8]. For dust thermal emission, we use the dust absorption coefficient per unit mass density (which includes both gas and dust) and unity length, the opacity κ_ν, which is related to the optical depth through

$$\tau_\nu = \kappa_\nu \int \rho \, {\rm d}s \tag{3.81}$$

Dust opacity is empirically approximated by a power law of frequency with exponent β (typically 1-2)

$$\left[\frac{\kappa_\nu}{\rm cm^2\,g^{-1}}\right] = 0.1 \left[\frac{\nu}{1000\,\rm GHz}\right]^\beta \tag{3.82}$$

8 It is essentially a blackbody contribution multiplying an efficiency term.

We can assume optically thin emission at millimetric and sub-millimetric wavelengths, so that the RJ approximation holds and we can write the radiative flux as

$$F_\nu = \frac{2k\nu^2}{c^2} T_d \tau_\nu \Omega_S \propto \nu^{2+\beta} \tag{3.83}$$

where $2+\beta$ is sometimes called *spectral index*. Measuring the flux at different wavelengths or frequencies we're able to obtain a curve known as *spectral energy distribution* (SED).

If you were to look at the SED for dust emission, you'd find out that a grey body fits rather well most of the observed profile, but with a significant deviation from expectations in the short-wavelengths range (mid/near-IR). Such "flux excess" is caused by the presence of Polyciclic Aromatic Hydrocarbons (PAHs) in the ISM.

The mass of the dust in the ISM can be calculated applying Mie's theory of absorption/scattering under "reasonable assumptions" (as an old professor of mine used to say). In fact, the monochromatic luminosity of a single (spherical) dust grain can be written as

$$L_\nu = 4\pi a^2 \pi B_\nu(T_d) Q_\nu \tag{3.84}$$

where Q_ν is an emission efficiency factor, that accounts for deviations of the dust grain from black body behavior.

If the total mass of dust in the cloud is M_d, while the mass of the single grain is m_d, the number of grains in the cloud is simply

$$N_d = \frac{M_d}{m_d} = \frac{3M_d}{4\pi a^3 \rho_d} \tag{3.85}$$

Now, if the emission is *optically thin*, the flux measured at a distance D is

$$F_\nu = \frac{N_d L_\nu}{4\pi D^2} \tag{3.86}$$

Hence, substituting all the needed quantities, we're able to find an expression for the total mass of dust in a cloud

$$M_d = \frac{4a\rho_d F_\nu D^2}{3B_\nu(T_d) Q_\nu} \tag{3.87}$$

ABOUT ME

Hi there, I'm Giacomo, sometimes known as Cael by a very small niche of people, the author of the notes you just finished reading.

I'm a Batchelor-graduate student at the University of Pisa now in the process of undertaking a major (Master degree) in Astronomy and Astrophysics at the same university. In my (meager) free time I'm an avid reader, gamer, anime-enthusiast as well as an independent author in the very process of publishing his first novel.

For those of you that may be interested I'll leave both the link to the novel's page as well to my instagram profile in the QRcodes right below.

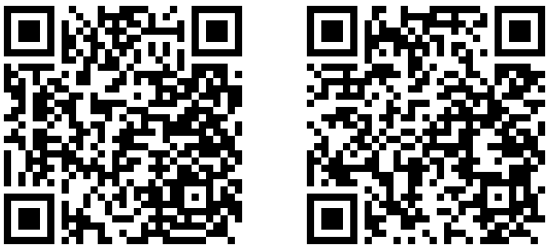


Figure 3.16: On the left is the link to my instagram profile, on the right the one to my novel's page (Python really has a library for anything, huh?).

Come say hi if you fancy action-fantasy stories with magic, swords, some comedy to garnish and a lot of other stuff :)

At last but not the least, I want to thank all my collegues and fellow students that helped (and hopefully will help me again) during the drafting of this text, pointing out errors and occasions for further expansion.

If you'd like your names to be included and take your fair share of credits, feel free to contact me and I'll add your name right away in the few lines that are left of this page.

I hope my understading of the complex interplayings going on in our Universe has managed to facilitate your own understading.

If not, well... oopsie.

See you, space cowboys...

Giacomo

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